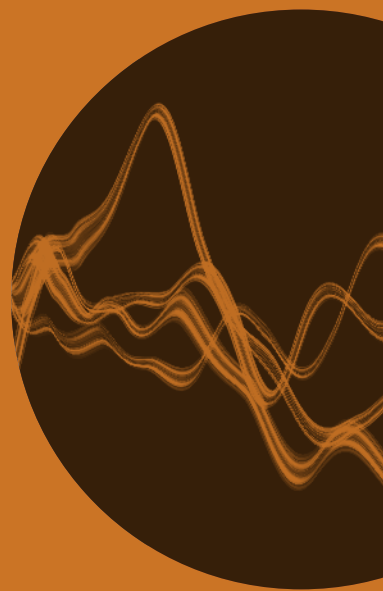
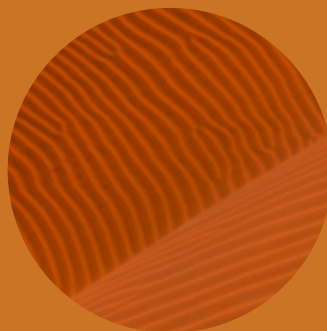


Review

ERICSSON
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CTO TECH TRENDS
CREATING INTELLIGENT
DIGITAL INFRASTRUCTURE

CRITICAL IoT
CONNECTIVITY
FOR INDUSTRY

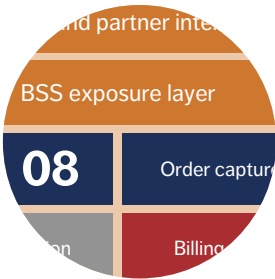
INTEGRATED ACCESS
AND BACKHAUL
IN 5G NR NETWORKS





08 5G BSS: EVOLVING BSS TO FIT THE 5G ECONOMY

Managing complex IoT value chains and supporting new business models requires more sophisticated business support systems (BSS) than those that communication service providers have used in the past. 5G-evolved BSS enable smooth collaboration between connectivity providers, service creators, partners, suppliers and others.



20 OPTIMIZING UICC MODULES FOR IoT APPLICATIONS

The ability to deliver low-cost Internet of Things (IoT) devices on a mass scale is at risk of being hampered by the high cost of the universal integrated circuit cards (UICC) currently required to provide connectivity. Until a less costly alternative becomes available, the IoT requires workarounds that either lower device cost or justify the price of UICCs by leveraging more of their capabilities.



30 FEATURE ARTICLE

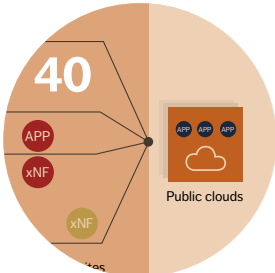
Future network trends: Creating intelligent digital infrastructure

The vision of a fully digitalized, automated and programmable world of connected humans, machines, things and places is well on its way to becoming a reality. In his annual technology trends article, our CTO Erik Ekudden explains the seven technology trends that are most relevant to the network platform's evolution to become the platform for innovation to meet any societal or industrial need.



40 THE FUTURE OF CLOUD COMPUTING: HIGHLY DISTRIBUTED WITH HETEROGENEOUS HARDWARE

Cloud computing is being shaped by the combination of the growing popularity of distributed solutions and increased reliance on heterogeneous hardware capabilities. As the role of distributed computing in cloud computing continues to expand, network operators, who have large, distributed systems already in place, have a golden opportunity to become major cloud players.



52 CRITICAL IoT CONNECTIVITY – IDEAL FOR TIME-CRITICAL INDUSTRIAL COMMUNICATIONS

Critical IoT connectivity is ideal for a wide range of Internet of Things use cases across most industry verticals. Mobile network operators are uniquely positioned to address the time-critical communication needs of individual users, enterprises and public institutions by leveraging their existing assets and new technologies in a systematic fashion.



64 INTEGRATED ACCESS AND BACKHAUL – A NEW TYPE OF WIRELESS BACKHAUL IN 5G

Integrated access and backhaul (IAB) is an advanced concept in 5G that shows significant promise in addressing the challenge of wireless backhaul of street sites. IAB has several advantages compared with other backhaul technologies, and if used properly, it could become an essential backhaul solution for 5G NR networks.



Ericsson Technology Review brings you insights into some of the key emerging innovations that are shaping the future of ICT. Our aim is to encourage an open discussion about the potential, practicalities, and benefits of a wide range of technical developments, and provide insight into what the future has to offer.

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Future network trends:
Creating intelligent digital infrastructure
by Erik Ekudden

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BRIDGING THE GAP BETWEEN PHYSICAL AND DIGITAL REALITIES

■ **THE KEY ROLE that connectivity plays in our daily lives has never been more obvious – not only for each of us as individuals but also for countless enterprises around the globe. Thankfully, despite the sudden, dramatic changes in our behavior in early 2020, networks all around the world have proven to be highly resilient.**

At Ericsson, we're committed to ensuring that the network platform continues to improve its ability to meet the full range of societal needs as well as supporting enterprises to stay competitive in the long term. The ability to bridge distances and make it easier to efficiently meet needs in terms of resource utilization, collaboration, competence transfer, status verification, privacy protection, security and safety is of utmost importance. Greater agility and speed will be essential.

My 2020 technology trends article, on page 30 of this issue of the magazine, explains my view of the ongoing evolution of the network platform in terms of the key needs that are driving its evolution and the emerging capabilities that will meet both those and other needs.

The first three trends all relate to bridging the gap between physical reality and the digital realm – that is, delivering sensory experiences and utilizing digital representations to make the physical world fully programmable. The emerging capabilities that I have highlighted this year are non-limiting connectivity, pervasive network compute fabric, trustworthy infrastructure and cognitive networks.

●● GREATER AGILITY AND SPEED WILL BE ESSENTIAL ●●

All seven of these trends serve as a cornerstone in the development of a common Ericsson vision of what future networks will provide, and what sort of technology evolution will be required to get there.

This issue of the magazine also includes five additional articles highlighting some of our latest research in the areas of cloud computing, the Internet of Things (IoT) and 5G advancements. The cloud computing article is particularly noteworthy, as it explains how we think network operators can best manage the complexity of future cloud deployments and overcome technical challenges.

The first IoT article in this issue explains how critical IoT connectivity can be used to address time-critical needs in areas such as industrial control, mobility automation, remote control and real-time media, while the second one tackles the challenge that today's universal integrated circuit cards (UICC) present to IoT growth.

With regard to 5G advancements, our BSS article explores how 5G-evolved BSS can help communication service providers transform themselves from traditional network developers to service enablers and ultimately service creators.

Another exciting 5G advancement that we present in this issue is integrated access and backhaul (IAB), an innovative concept that shows significant promise in addressing the challenge of wireless backhaul of street sites.

We hope you enjoy this issue of our magazine and we'd be delighted if you share it with your colleagues and business partners. You can find both PDF and HTML versions of all the articles at: www.ericsson.com/ericsson-technology-review



Erik Ekudden

ERIK EKUDDEN

SENIOR VICE PRESIDENT,
CHIEF TECHNOLOGY OFFICER AND
HEAD OF GROUP FUNCTION TECHNOLOGY

5G BSS:

Evolving BSS to fit the 5G economy

5G offers communication service providers an unprecedented opportunity to enhance their position in the value chain and tap into new revenue streams in a variety of industry verticals. A successful transition will require business support systems (BSS) that are evolved to fit the 5G economy.

JAN FRIMAN,
MICHAEL NILSSON,
ELISABETH MUELLER

The rapidly expanding Internet of Things (IoT) and all the new capabilities available in 5G have opened up a wealth of opportunities for communication service providers (CSPs) beyond their traditional markets, particularly in verticals such as automotive, health care, agriculture, energy and manufacturing. To monetize them, CSPs will need to meet the expectations of a broader range of stakeholders and be able to handle complex ecosystems.

One of the primary roles of business support systems (BSS) is to manage a CSP's relationships with its stakeholders by keeping track of agreements, handling orders, generating reports, sending invoices and so on. In the past, these stakeholders were generally limited to consumers, resellers, partners and suppliers. In the 5G/IoT business context, though, more complex ecosystems are arising that BSS must evolve to support. To do so, the requirements of a larger, more diverse group of stakeholders must be taken

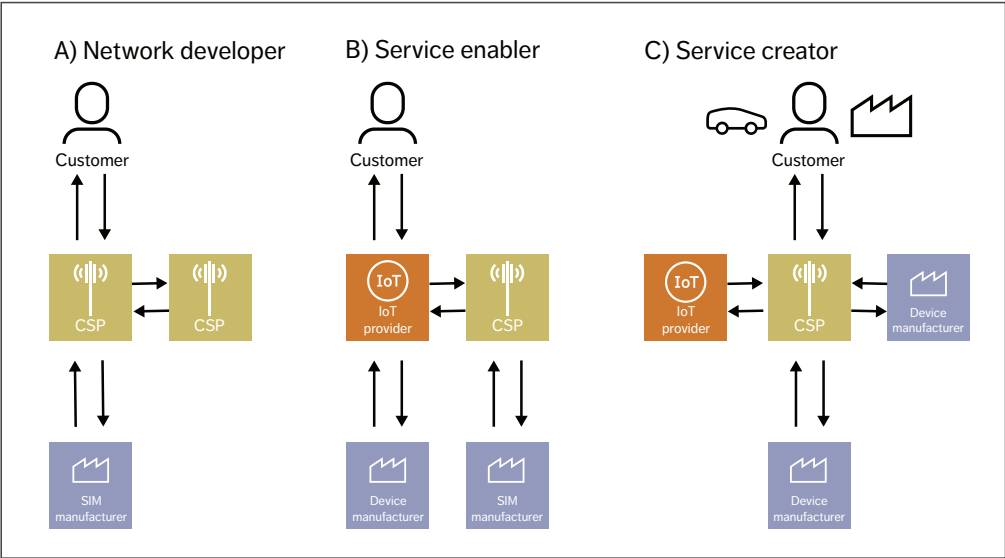


Figure 1 The evolving role of the CSP in the IoT ecosystem

into account, and mechanisms must be established to manage the relationships between them. Examples of new stakeholder groups that need to be considered in the 5G/IoT business context include:

- » Enterprises and industry verticals that require solutions beyond telecoms
- » New types of suppliers such as IoT device providers and suppliers of eSIM (embedded SIM) and related technologies
- » Platform providers that specialize in specific IoT or edge clusters or groups of use cases such as massive and broadband IoT platforms, industrial IoT platforms and content data networks
- » Integrators that specialize in specific verticals such as asset management, mission-critical services or automotive that combine capabilities from multiple stakeholders to address consumer needs.

Network developer, service enabler or service creator?

Looking ahead, the capabilities that a CSP needs in its BSS solution will depend on the role it plays – or aims to play – in the IoT ecosystem. Figure 1 illustrates the three role types: network developer, service enabler and service creator.

In the traditional network developer role, a CSP acts solely as a cellular connectivity provider by offering solutions such as radio, core network and communication services. In this role, the CSP's business models are consumer focused. Its role in the IoT ecosystem is limited.

In the service enabler role, the CSP extends its services by incorporating additional capabilities such as cloud/edge and IoT enablement and shifts focus to business customers and industry verticals. The CSP becomes a service enabler for 5G and the IoT, acting as a supplier of connectivity and platform services. As a service enabler, the CSP's business

models are extended to business-customer focused with respect to 5G IoT.

In the service creator role, the CSP transitions from being a connectivity and platform provider to creating new digital services and collaborating beyond telecoms to establish digital value systems. As a service creator, the CSP partners with suppliers to deliver new services all the way up to full IoT solutions, taking on the roles of integrator, distributor or co-seller.

BSS for all three CSP roles

Traditional BSS support the CSP in the network developer role, in which the CSP charges for voice, text and data services based on consumption or subscription level. The main requirements for these BSS are:

- » Customer management, traditional partner business (roaming partners), charging and billing, and finance modules
- » Order capture and order execution for new telco subscriptions and/or add-on offerings
- » Charging and balance/quota management in BSS, as well as mediation
- » Interaction with operations support systems (OSS) for network provisioning.

Evolving BSS to support a CSP in a service enabler role requires a shift in focus to the needs of enterprise customers and IoT use cases. The BSS must be transformed into a system that is able to monetize IoT/5G platforms and edge deployments, which requires significant changes in both the functional and non-functional space. In the non-functional space, this mainly involves scalability

– that is, enabling the BSS to handle traffic and a large number of devices at IoT scale.

In terms of functionality, the BSS enhancements required by service enablers include:

- » Automation of full life-cycle management for devices/IoT resources supported by flexible orchestration, including exposure of services for managing relationships with business customers
- » Support for batch orchestration, flexible supply agreements and contracts for non-telco services with associated charging models
- » Service exposure of network capabilities, so that IoT providers can bundle their offerings with connectivity and sell them on to their customers
- » Service exposure of BSS and OSS capabilities to enable efficient ordering processes, especially with regard to the management of mass subscriptions.

Supporting a CSP in the service creator role, where the full life cycle of partners must be taken into account, requires BSS with further functional extensions. The stakeholder ecosystem of service creators is significantly more complex, as the customer base broadens to include verticals and the CSP starts offering full solutions beyond telecoms. As a result, BSS for service creators must include extensive and flexible partner relationship management. Supply chain management is especially important.

The capacity to expose network capability as well as BSS and OSS capabilities is critically important to a CSP’s ability to deliver on service creation beyond

Terms and abbreviations

API – Application Programming Interface | **BSS** – Business Support Systems | **CSP** – Communication Service Provider | **IoT** – Internet of Things | **ODA** – Open Digital Architecture | **OSS** – Operations Support Systems | **SBI** – Service-Based Interface | **SDK** – Software Development Kit | **SLA** – Service Level Agreement

BSS solution level	Capabilities
5G enabled	• 5G service-based interface (SBI) support (charging function) • Network slicing support in BSS and OSS • Classic roaming partners • Containerization and microservices • Common technology stack
IoT and edge monetization	• ID management and correlation • Life-cycle management for IoT devices • Business customer and 5G/IoT enterprise management • Charging in multilevel hierarchies • Supply agreements • Flexible orchestration of ordering processes • Service exposure for device management • Open API exposure • Continuous integration/continuous delivery (CI/CD) for service exposure • Enterprise self-care • Multiparty charging and best-effort charging • Private networks • Platform partnerships • Contract for non-telco services (IoT/edge enabled) • Charging models for non-telco services • Multi-tenancy • Charging and billing on behalf of • Location-aware services • Blockchain for smart contracting • Service Level Agreement (SLA) management
Full 5G ecosystem	• Partner relationship management • Partner catalog • Partner revenue sharing • Reconciliation and settlement • Flexible billing • Platform as a service and distributed cloud • Edge platform services • Multi-access edge computing (MEC) • BSS as a service • Continuous monitoring • Artificial intelligence and machine-learning automation • CI/CD

Table 1 Key capabilities of the three BSS solution levels

telecoms, so that partners can develop tailored applications and deploy them on the operator’s infrastructure.

Finally, the new business models available to CSPs as service creators require new monetization models for charging and billing. For example, multiparty charging, revenue sharing and profit

sharing all require extended billing and reconciliation functionality.

BSS solution levels and key capabilities

Table 1 organizes and sequences key BSS capabilities based on technical dependencies and/or level of complexity. One by one, these capabilities

SCALABILITY ALONE IS NOT ENOUGH TO HANDLE MASSIVE AMOUNTS OF DEVICES

add on to each other, continuously increasing BSS maturity and transforming the BSS into a system capable of supporting all the new use cases and business models that characterize the 5G/IoT ecosystem.

The first evolution step – ‘5G enabled’ in Table 1 – provides support for new 5G standards and concepts, which enables a drastic increase in data transmission throughput while maintaining focus on traditional consumers. Applying containerization and a common technology stack will assure the scalability of the BSS solution to meet the increased throughput demands of the network.

At the next solution level, IoT and edge monetization, the focus shifts to business customers. These new capabilities enable the CSP to provide extended support for enterprises when it comes to 5G and IoT use cases by covering IoT device management, support for non-telco service charging and multi-party charging as well as IoT and/or edge-platform monetization. In addition, service exposure enables self-service for enterprises along with application development for the optimization of IoT devices. The number of 5G/IoT use cases that the CSP is able to support increases drastically at this stage.

The addition of partner capabilities at the full 5G ecosystem level allows the CSP to address totally new customer segments beyond telecoms and provide industry-specific solutions to verticals. A CSP can create new services (even deliver BSS as a service), and offer these services on a marketplace to reach new segments of business customers. The multitude of partnerships require support for new business models that allow flexible charging, revenue sharing and billing.

5G reference architecture for BSS

From a high-level architectural viewpoint, BSS in the 5G/IoT ecosystem closely resemble traditional

BSS, with similar interfaces to surrounding systems. The BSS architecture in Figure 2 is presented in the Open Digital Architecture format [1]. It is divided into party management, core commerce management, intelligence management, production and engagement management. Production includes the southbound application programming interface (API) layer to the network infrastructure, IoT platforms, cloud/edge and OSS, while engagement management includes the northbound API layer to customers and partners.

5G and the IoT place several challenging requirements on new capabilities in the BSS architecture that are not directly visible at a high level. All functional areas are affected by the 5G evolution and are extended to support the new requirements and possibilities, most notably in the areas of mass-device management, device and resource life-cycle management, subscription management, charging models for non-telco services and multiparty charging.

IoT-scale mass-device management

The sheer number of connected devices in the 5G/IoT world is a major challenge for BSS to manage. While current BSS architectures are scalable, they will be too costly for IoT use cases due to the large data footprint and processing need of each device. Scalability alone is not enough to handle massive amounts of devices. To address this, 5G-evolved BSS must have a persistence and management model that is lightweight enough to allow a large number of devices to use the same footprint as one traditional device. This can be addressed using concepts such as herding, where each individual device only requires a minimal data footprint. The behavior of each individual device is determined by the herd configuration, which is a single specification per herd.

Life-cycle management of IoT devices and resources

Managing the life cycles of IoT devices and resources is another significant challenge for BSS. In many emerging IoT applications, the ability to

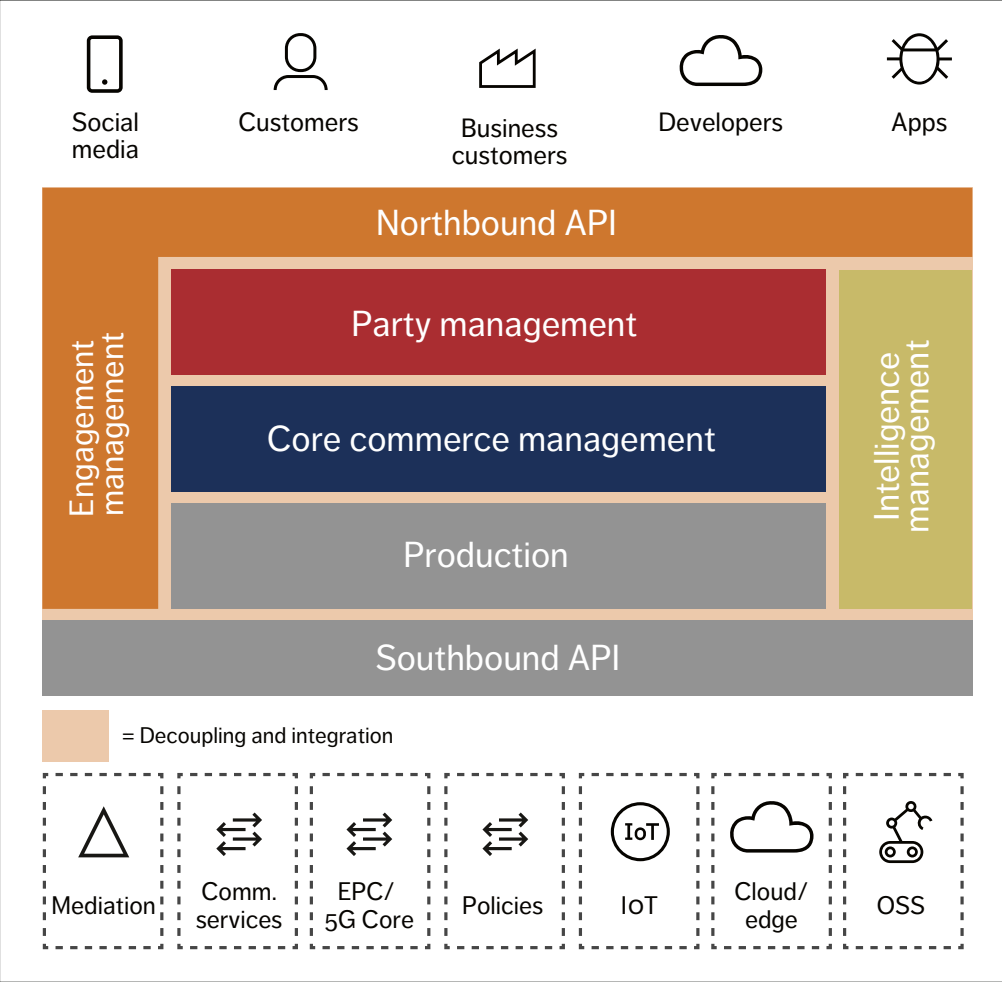


Figure 2 5G reference architecture for BSS

monitor the state of the device throughout its life cycle is not sufficient. For example, contracts that cover large herds of devices are likely to be based on recurring charges per active device. In these scenarios, the aggregated numbers of devices per state become key parameters in the calculation of charges.

The calculation of charges related to IoT devices is also complicated by the fact that the state of the device can influence the charged party. One example of this challenge is IoT devices that are mounted in vehicles at a factory. The factory personnel will likely want to test that the device is working before shipping the vehicle to the reseller. The reseller may

●● ONE EFFECTIVE APPROACH IS TO EXPOSE APIs AND TOOLS THAT ALLOW PARTNERS ... TO ONBOARD AND MANAGE DEVICES ●●

then want to demonstrate the service the device provides to prospective buyers, before a consumer ultimately buys the vehicle and starts using the service. At each of these stages, the charged party and charging model may be different depending on the state of the device. Overcoming such challenges requires a BSS architecture that can provide up-to-date state information per individual device or resource as well as aggregated information to the rating, charging and billing functions.

Subscription management for IoT devices

Subscription management is another area that must evolve to fit the new 5G/IoT business context. Traditional BSS are built to manage consumer subscriptions. They are not capable of handling the massive number of devices in IoT use cases in a cost-efficient manner. Subscription management in 5G-evolved BSS requires a high level of automation and solutions that reduce the processing footprint to onboard and manage devices, services and products. One effective approach is to expose APIs and tools that allow partners or even consumers to onboard and manage devices.

To gain efficiency and minimize management, pools of services and products can be linked to herds of devices, instead of applying individual services to device relationships, which is the common practice in BSS today. The service instances linked to herds are kept to a minimal footprint and the majority of the parameters needed for processing can be kept on specification level. This change will enable more efficient processing in BSS and reduce the number of scenarios that require mass provisioning.

Unlike traditional BSS, 5G-evolved BSS must be able to capture and create the network charging data records (charging function). This task provides the

BSS with a unique opportunity to determine which charging, balance management and aggregation functions must be performed, and use this knowledge to monetize the usage of the 5G network. For instance, the BSS can monitor allowances and balances in real time, if so required by a partner agreement, or decide to postpone the rating and balance management to a near real-time asynchronous flow.

Allowing the BSS to decide the importance and risk level of each event based on agreements, Service Level Agreements (SLAs) and operator business rules makes it possible to accommodate multiple charging models simultaneously. Among other things, this approach enables real-time monitoring of individual device herds, while at the same time providing partner ratings for one or multiple involved partners in a continuous, near real-time, flow for individual device sessions.

Charging models for non-telco services

5G-evolved BSS must also support the management and monetization of services that are not traditional telco services, such as those for the IoT platform or application hosting at the edge. In the past, BSS have traditionally relied on a well-defined set of parameters provided through standardized protocols, but this approach will not be sufficient when entering the non-telco service arena. To monetize on non-telco services, the 5G-evolved BSS must have the flexibility to use previously unknown identifiers and parameters, especially in the charging and billing systems.

The usage of a non-telco service can be monetized using something as simple as a network slice identifier to determine how to aggregate and charge for a service. In other instances, a much more complex model must be used, involving multiple input parameters for each event to determine which party or parties should be charged and which charging model should be applied. Consequently, the charging and billing solution in 5G-evolved BSS must provide the flexibility to map and evaluate non-telco identifiers and other parameters at configuration time.

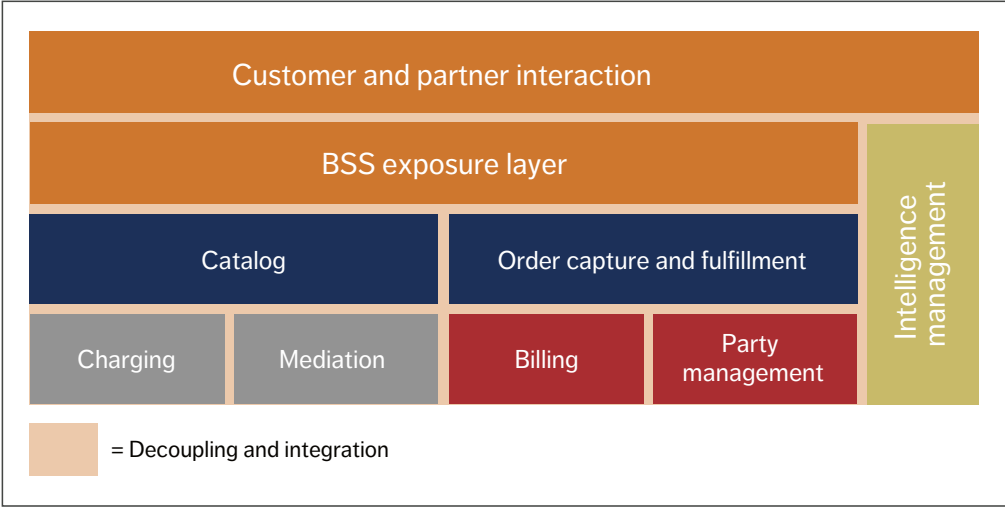


Figure 3 Ericsson's digital BSS implementation architecture

Multiparty charging

While traditional BSS are able to handle roaming partners and wholesale agreements, they are not equipped to handle the dramatic increase in different types of partner agreements in the 5G/IoT ecosystem. The ability to handle a wide variety of partner agreements and support the onboarding of partners and related charging models will be crucial to CSPs' ability to monetize on expected IoT growth and avoid becoming bit-pipe wholesalers.

In the 5G/IoT ecosystem, a single event that BSS receive from the 5G core network can trigger a complex value chain that requires multiple parties to be charged or share revenue. A CSP cannot rely on traditional techniques to handle this complexity – doing so would mean postponing charging or revenue share distribution until the bill run.

To deliver up-to-date information to the relevant partners, the CSP needs BSS that can process the entire value chain as soon as any activity occurs that impacts them. This does not mean that everything must be processed in real time, but rather that events must be handled in an online asynchronous process. For example, when BSS grant consumers the right to

access specific services, the event is followed up by a post-session process to calculate and distribute the charges/revenue share for the involved partners. As a result, the relevant partners have access to up-to-date information within seconds, rather than at the end of the day or at the bill run as they would in traditional BSS.

In 5G-evolved BSS, different events for the same service can have different charge or revenue share distribution. One-time fees, recurring charges or usage fees can all have different distribution rules and include one or more partners. For example, it is possible for an operator to charge a one-time fee to a consumer and keep all of the revenue, while also charging a recurring fee to the same consumer and splitting that revenue with a partner that provides the consumer device on a rental basis.

Digital BSS architecture for 5G and the IoT

Figure 3 shows the key components of Ericsson's digital BSS architecture. The color scheme indicates the relationship between the components in this architecture and the functional ODA architecture shown in Figure 2.

The front-end channels in the customer and partner interaction layer and the BSS exposure layer are deployed as a microservice architecture to facilitate business agility, scaling and the introduction of customized solutions as per operator needs.

Further down in the stack, the architecture is based on mini services, primarily to optimize footprint, performance and latency.

Table 2 maps out the 5G evolution areas in BSS to the main functional blocks in our digital BSS

BSS functional block	5G evolution areas
Customer and partner interaction	• Catalog driven, omnichannel • B2C and B2B digital front end: customer/partner journeys • B2C and B2B CPQ (configure, price and quote), frame contracts • B2B2X marketplace
BSS exposure layer	• OpenAPI exposure • Loosely coupled principle • SDK to support API aggregation
Catalog	• Exposure for partner product creation • Enhanced bundling with partner products • Product models for network resources • Product models for enterprise products • Partner catalog • Multi-device offerings
Order capture and fulfillment	• Ecosystem orchestration • New business model support
Charging	• Support for new charging trigger points • Manage communication services at IoT scale • Charging life-cycle management as a part of mass IoT device and mass subscription life-cycle management • Multiparty charging • Charging in hierarchies • Charging on behalf of • Non-telco service charge
Mediation	• Call detail record generation for 5G • Online mediation SBI -> diameter
Party management	• Extended B2B (supply agreements, non-telco contracts) • Digital partner management
Intelligence management	• SLA management • Data lake
Billing	• Life-cycle management as a part of mass IoT device and mass subscription life-cycle management • Multiparty billing • Billing on behalf of • Revenue sharing • IoT partner settlements

Table 2 Prioritized 5G evolution areas in the main BSS functional blocks

architecture. Containerization, microservices and a common technology stack are common to all blocks.

CSPs NEED BSS THAT ARE EVOLVED TO MANAGE COMPLEX VALUE CHAINS

Conclusion

The 5G network evolution presents communication service providers with the opportunity to transform themselves from traditional network developers to service enablers for 5G and the Internet of Things, and ultimately to service creators with the ability to collaborate beyond telecoms and establish lucrative digital value systems. Along the way, this journey opens up substantial new revenue streams in verticals such as industrial automation, security, health care and automotive. To successfully capitalize on this opportunity, CSPs need BSS that are evolved to manage complex value chains and support new business models.

5G-evolved BSS enable smooth collaboration between connectivity providers, service creators, partners, suppliers and others that results in the efficient creation of attractive and cost-effective services. Optimized information models and a high degree of automation are required to handle huge numbers of devices through open interfaces. Deployment in a cloud-native architecture ensures flexibility and scalability. It is important to keep the business logic, interfaces and information models of 5G-evolved BSS flexible, so they can be adjusted to suit the value chains and business models of the different industry verticals.

At Ericsson, we will continue to evolve our BSS offering to support our customers on their journeys from network developers to service enablers, from service enablers to service creators and beyond. As part of this work, we are also firmly committed to driving and contributing to relevant standards in the BSS area and participating in open source and developer communities to promote openness and interoperability.

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OPTIMIZING

UICC modules for IoT applications

The UICCs used in all cellular devices today are complex and powerful minicomputers capable of much more than most Internet of Things (IoT) applications require. Until a simpler and less costly alternative becomes available, it makes sense to find ways to reduce the complexity of using them and use their excess capacity for additional value generation.

BENEDEK KOVÁCS,
ZSOLT VAJTA,
ZSIGMOND PAP

UICCs are used today to facilitate network connection in all 3GPP user equipment – mobile phones, IoT devices and so on.

■ The most important tasks of UICC modules – commonly referred to as SIM cards – in today’s mobile networks are to store network credentials and to run network security and access applications in a secure and trusted environment. In addition, they are also capable of storing a large amount of extra information and running multiple toolkit applications. A UICC’s own operating system provides a full Java environment. It can run

dozens of Java-based applications in parallel and support powerful remote management operations. Backward-compatibility is provided by running a network service application on UICC modules, which can emulate the file system for storing necessary credentials and old-school smartcard protocols, extended with features such as enhanced security, extended telephone register and operator logo image. The interface between the UICC module and the user equipment (devices) is standardized, which enables operators to run value-added applications, such as mobile wallet or mobile lottery, on the UICC module.

While the advanced features of UICC modules continue to provide considerable value in mobile phone applications, most of them are superfluous in IoT applications. In light of this, the industry is working to find a less sophisticated solution that is more appropriate for applications that require massive numbers of devices in price-sensitive environments. Industry alignment on such a solution is expected to be a challenging and time-consuming process, however, due to the fact that the IoT area is fragmented into many different verticals, application areas and use cases.

Ericsson is fully committed to supporting the long-term, industry-aligned solution. In the meantime, however, it is vital to find workarounds to ensure that the cost of UICCs does not stifle IoT growth. While the definitive solution to the question of what should replace the UICC is hard to predict, two mid-term workarounds are clear: the complexity of using UICCs and leveraging their excess capacity to generate additional value.

Reducing the complexity of using UICCs

There are three main approaches to reducing the complexity of using UICCs in IoT applications: optimization, usage of 3GPP standardized certificate-based authentication, and virtualization.

Optimization

A typical operator profile on a 3GPP consumer mobile phone is up to tens of kilobytes; the average IoT sensor only requires 200-300 bytes. And of all the functionality that a UICC can provide, an IoT device only really needs the Universal Subscriber Identity Module application and the remote SIM provisioning (RSP) application, which allows remote provisioning of subscriber credentials (also known as operator profiles).

One good way to significantly reduce the footprint of the UICC is to optimize the operator profile and the necessary software environment within the UICC module. Doing so not only saves storage in the device but also reduces energy consumption during over-the-air download. Further size reduction of the device may be achieved when the UICC is completely integrated into the baseband modem or application processor (integrated UICC or iUICC [2]). This simplified and integrated solution could work effectively for use cases that require low-cost, simple, secure and low-power IoT devices in high volumes.

The use of an iUICC requires an effective RSP protocol [3, 4] that makes it possible to change subscription credentials. Current RSP standards are too complex for iUICCs for many reasons, including their use of HTTPS

Definition of key terms

Identity describes the link between the identifier of an entity and the credentials that it uses to prove that it is the rightful owner of the identity.

First used in Finland in 1991, the original **subscriber identity module (SIM)** was a smart card with a protected file system that stored cellular network parameters. It was designed to connect expensive user equipment – mobile phones – with expensive subscriptions to the cellular network.

When it became clear that smart cards did not have the capacity to provide an adequate level of security in next-generation cellular networks, they were replaced with **universal integrated circuit cards (UICCs)** – minicomputers equipped with general microprocessors, memory and strong cryptographic co-processors [1].

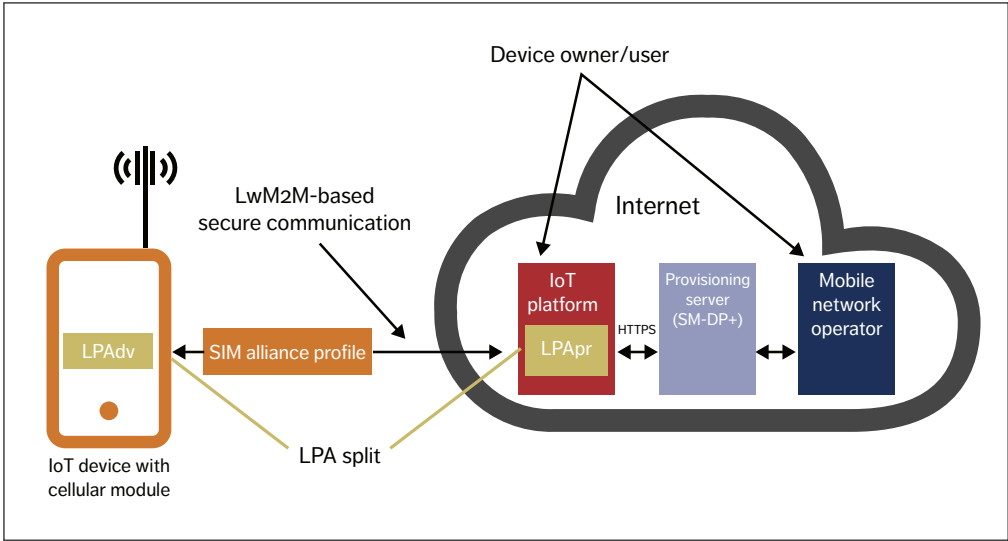


Figure 1 Remote provisioning using IoT-optimized technology

(Hypertext Transfer Protocol Secure) and reliance on SMS support. HTTPS is typically not part of the protocol stack of constrained low-power IoT devices. Instead, these devices use a stack with Constrained Application Protocol (CoAP), Datagram Transport Layer Security (DTLS) and User Datagram Protocol. In some cases, the Lightweight Machine-to-Machine (LwM2M) protocol is used on top of CoAP for device and application data management. The use of only one stack keeps the cost of the device down.

Ericsson proposes utilizing the same protocol stack for profile download and profile management as is used for device and application data management. **Figure 1** illustrates how to achieve this by adapting the GSMA embedded-SIM solution for consumer devices for use with IoT devices. In this solution, the local profile assistant (LPA) is split into two parts. To reduce device footprint, the main part of the LPA (including the use of HTTPS) is moved from the device to a device

or connectivity management server. The device management protocol stack (Open Mobile Alliance (OMA) LwM2M [1], for example) handles the communication between the two LPA parts. Profile protection is still end-to-end between the iUICC/embedded-UICC (eUICC) and the provisioning server (Subscription Manager-Data Preparation – SM-DP+).

Usage of 3GPP standardized certificate-based authentication

Another way to reduce the need for a UICC is to use a network authentication mechanism different to the classical 3GPP Authentication and Key Agreement (AKA). The use of certificates is a classic solution used in the internet that may easily fit into the existing network architecture of an enterprise/service provider. In public 5G networks, authenticating with certificates is possible as a secondary authentication for a service using AKA, but only after primary network

authentication has been performed. According to the 3GPP, authentication in private networks such as Industry 4.0 solutions may rely entirely on certificate-based solutions such as Extensible Authentication Protocol over Transport Layer Security. Without a UICC for securely storing and operating on secret long-term credentials for network access authentication, another secure environment with secure storage solution is needed.

For certain applications a lower level of security might be accepted. The value of the data that the IoT device provides or handles, in relation to the cost of the IoT device, determines the required security level of the secure environment for protecting network access authentication credentials. In the case of a UICC being used, it determines the realization of the UICC functionality. For some low-cost constrained IoT devices, a realization using a hardware-isolation-based trusted execution environment may be acceptable. As there is no universal and perfect solution, operators must decide which solution is most suitable for any given application. It is likely that the UICCs and eUICC-based solutions will remain the technology of choice in public networks for the next few years.

Virtualization

Virtualizing the UICC is yet another alternative that addresses the cost issue associated with UICC technology. One way to do this is to run a UICC environment in a virtual machine (or at least on a separated processor core) inside the application processor or the baseband modem. Another approach is to store the operator profiles in the security zone of the application processor, then download them to empty physical UICC hardware on demand.

The biggest advantages of these virtualization solutions is flexibility and better utilization of existing hardware resources, while at the same time maintaining many of the advantages of current technology. These methods are particularly effective

when an IoT device needs to manage multiple operator profiles – a circumstance that will become increasingly common, according to an analysis carried out by the GSMA [5].

The disadvantages of virtualization are similar to those of certification-based solutions. Most notably, certification is harder when a trusted environment is integrated with the rest of the device compared with using an isolated UICC or eUICC.

OPERATORS MUST DECIDE WHICH SOLUTION IS MOST SUITABLE FOR ANY GIVEN APPLICATION

Generating additional value from the UICC

Experience shows that it is significantly less expensive to limit a protected and certified manufacturing environment to a dedicated hardware module such as a UICC than to ensure that all the software running in the mobile equipment can be trusted. In light of this, we believe that communication service providers will continue using UICC modules for at least the next 5-10 years. During this period, it makes sense to exploit the potential of the UICCs to better support IoT applications by creating value-added services for operators and enterprises. Three examples of this are using the UICC as cryptographic storage, using it to run higher-layer protocol stacks, and using it as a supervisory entity.

Using the UICC as cryptographic storage

UICC modules were designed to serve as cryptographic storage and are used today mainly for the storage of security credentials for 3GPP connectivity. We propose, in accordance with GSMA IoT SAFE [1], that the UICC itself should also be used as a crypto-safe for the IoT platform, providing support to establish encrypted connection of the applications.

A generic IoT device has multiple identities for use in multiple security domains. Every identity has at least one identifier and credential, all of which must be stored somewhere. Although there are multiple options, a hardware element that is powerful enough to play the role of the root of trust is definitely needed. The UICC is perfect for this role, as it is already used as an identity for 3GPP networks, storing International Mobile Subscriber Identity, intensified charge-coupled device, Wi-Fi and OMA LwM2M [6] credentials along with dozens of other identifiers. The necessary trusted and certified environment and infrastructure are already available to manufacture the module, download and update its content and carry out remote management as well.

To cover every aspect, UICC-based solutions require cooperation between the UICC ecosystem and the IoT device security subsystem (ARM Trust Zone [7], for example). ID and credential management itself is device-independent, which saves development cost and increases the security level. Additional advantages of using UICC as a root of trust are:

- » it has its own local processor
- » it is usually equipped with powerful cryptographic co-processors
- » it comes with a powerful, standardized remote management subsystem (RMS)
- » it is handled through a separate logistics chain.

The UICC can generate key-pairs and store private keys for multiple security domains effectively and securely. Effectiveness comes from its powerful cryptographic co-processors, while security is provided by the combination of the standardized RMS and the UICC’s ability to run cryptography processes inside the module. This means that the keys never leave the hardware and therefore they cannot be exposed to the application. Not only does this architecture provide security, it can also securely tie the 3GPP connectivity credentials and other IoT certificates to each other.

The recently released GSMA IoT SAFE [8] offers a solution where the UICC is utilized as a root of trust for IoT security. Here, an applet on the UICC/ eUICC provides cryptographic support and storage of credentials for establishing secure communication (for example, using DTLS) to an IoT service. The existing UICC management system (UICC.OTA mechanism) is used by the operator to establish trusted credentials between the device and the IoT service. The GSMA IoT SAFE defines an application programming interface for interoperability between SIM applets from different operators.

EFFECTIVENESS
COMES FROM ITS POWER-
FUL CRYPTOGRAPHIC
CO-PROCESSORS

Using the UICC to run
higher-layer protocol stacks

In addition to providing security and encryption functions, UICC modules could also serve as main application processors. Today, a low-cost, sensor-like IoT device usually has at least three processors on board: one is on the UICC module, another runs inside the baseband modem, and a third – the application processor itself (sometimes combined) – collects data and hosts higher level communication stacks such as LwM2M, CoAP or MQ Telemetry Transport.

Shifting the higher-level communication stack from the application processor to the UICC module can lead to cheaper hardware and lower development costs, as well as providing a unique approach to interoperability. As a result, some modem manufacturers have implemented these protocols inside the modem, running a complete OMA LwM2M protocol stack in the baseband chip, for example. While this may free up an external application processor and speed up device development, this solution is rather inflexible.

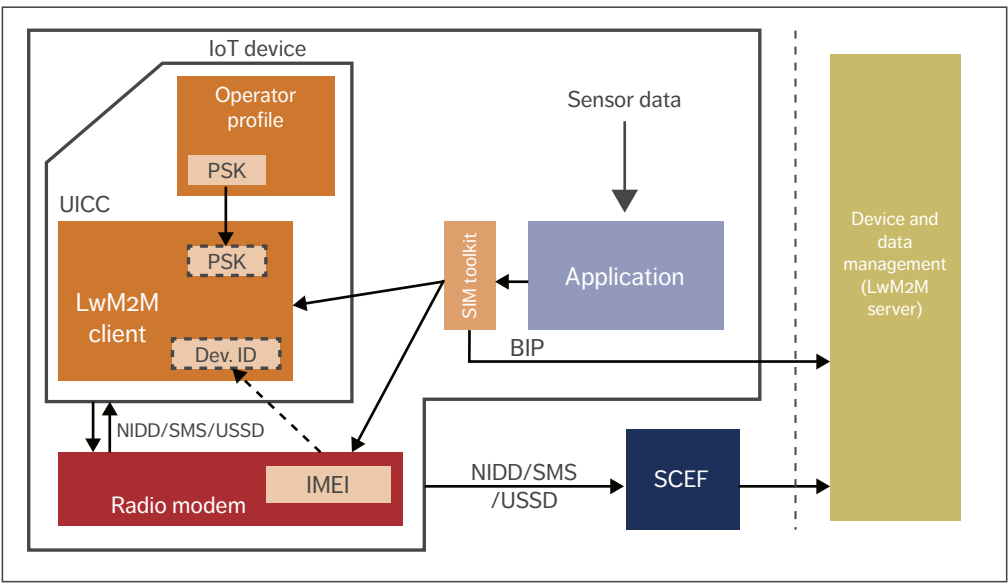


Figure 2 IoT device with LwM2M client running on the UICC module, using NIDD

Since modem firmware is a closed environment, it is difficult to upgrade and to customize its protocol stacks (extending them with proprietary added values). In addition, a small security hole in the protocol stack can be enough for a hacker to take control of the whole modem.

Alternatively, these higher-layer protocol stacks can be moved to the UICC. Figure 2 depicts a block diagram of a device, where the OMA LwM2M client runs on the UICC module and uses a non-IP data delivery (NIDD) protocol connection to send information to the device management system.

Running higher-level protocols in the UICC module can improve security in several ways. For example, it is possible to run the LwM2M stack over a NIDD connection [9] and even to allow this code to execute on the UICC module instead of on the device processor. In this scenario, command/control is never exposed on the

IP layer because it is running in the signaling network of the operator. An additional advantage of this approach is that it increases interoperability.

There is a standardized way of upgrading the communication stack in the UICC – it is even possible to insert the communication stack into the operator profile. This does not completely solve compatibility and interfacing problems, but a certified operator can handle these issues on a higher security level to provide wider solution matching.

In the simplest IoT devices, it might even be possible to run the actual IoT application on the UICC module. This would open for edge-computing solutions in which simple tasks are executed on the device – data filtering to reduce the amount of data being sent over the air, for example. Security can also be improved if the binary is stored on the UICC instead of on the device application processor.

Using the UICC as a supervisory entity

Zero-touch provisioning (ZTP) is yet another possibility for better utilization of the UICC module. ZTP refers to the possibility of adding an identity to a device when required, with automatic setup of the working environment (requiring manual intervention).

An effective automatic provisioning system requires remote provisioning management, key and credential storage, identity mapping of UICC modules and applications as well as strong flexibility in case of operator profiles, but all of this is far from enough. Provisioning of IoT devices is a complex, slow and costly procedure. Although there is a joint effort to extend mobile networks to support standardized, automatic device and subscription provisioning, it is at a very early stage.

A UICC MODULE CAN STORE SENSITIVE INFORMATION

During the provisioning procedure, two or more identities are given to the device, which entails that these identifiers are downloaded, and different subsystems are configured (mobile network, device

management system, data management system, and so on). Several standardized technologies exist to support this process but, unfortunately, they are not connected into a working, efficient, fully automated and cooperative system.

The most straightforward way to connect different subsystems in a flexible and programmable way is to run a centralized service above or at the same level as these subsystems. This ZTP service is connected to the 3GPP network (for instance to subscriber data management), to the SM-DP+ system (usually operated by the UICC module vendor or an independent bootstrap operator), to the device management system and to the data management system. The connection to the IoT device itself, to the manufacturer or even to the installer of the device can also be established. The main purpose of this service is to drive the IoT device through the steps of automatic device provisioning from the very beginning (ordering the device) to the final decommissioning.

Although this over-the-top service (OTT) can speed up the provisioning process significantly, it has some disadvantages. It should not store sensitive data, but only manage it indirectly. Furthermore, if the device has no connection at all, it cannot do anything. Scaling could also be a problem.

Terms and abbreviations

AKA – Authentication and Key Agreement | **BIP** – Bearer Independent Protocol | **CoAp** – Constrained Application Protocol | **DTLS** – Datagram Transport Layer Security | **eUICC** – Embedded UICC (soldered to the device board) | **HTTPS** – Hypertext Transfer Protocol Secure | **IMEI** – International Mobile Equipment Identity | **IoT** – Internet of Things | **IUICC** – Integrated UICC (integrated to a microchip) | **LPA** – Local Profile Assistant | **LPA_{adv}** – LPA (device), interfacing to the UICC | **LPA_{pr}** – LPA (proxy), interacting with the device owner and SM-DP+ | **LwM2M** – Lightweight Machine-to-Machine | **NIDD** – Non-IP Data Delivery | **OMA** – Open Mobile Alliance | **OTT** – Over-the-Top | **PSK** – Pre-shared Keys | **RMS** – Remote Management Subsystem | **RSP** – Remote SIM Provisioning (protocol) | **SCEF** – Service Capability Exposure Functions | **SM-DP** – Subscription Manager–Data Preparation | **UICC** – Universal Integrated Circuit Card | **USSD** – Unstructured Supplementary Service Data | **ZTP** – Zero-Touch Provisioning

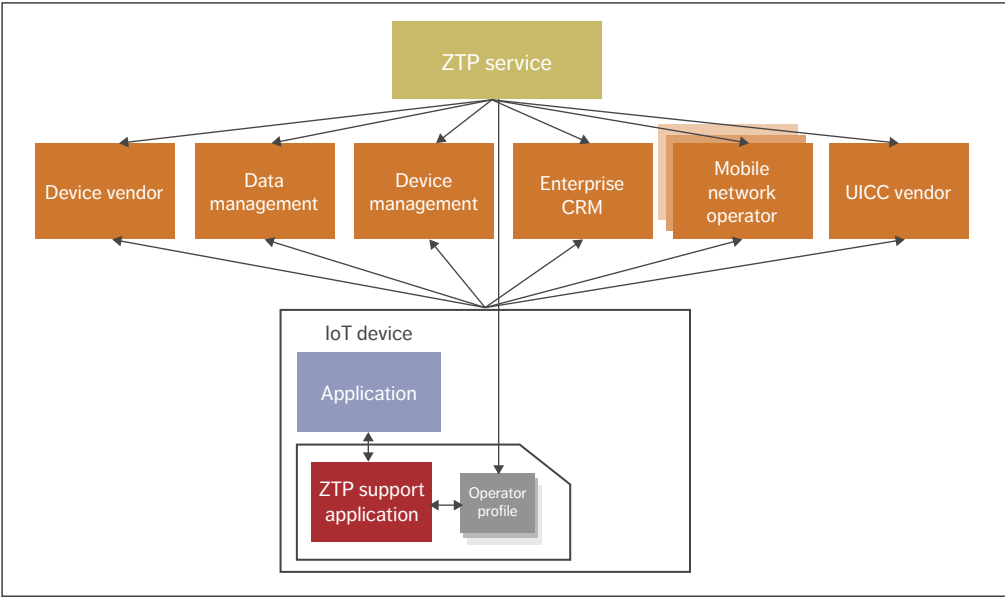


Figure 3 ZTP system with central ZTP service and UICC support

This is where a UICC application can help and support an OTT ZTP service. A UICC module can store sensitive information from different security domains. As it works close to the IoT device, it can do corrective actions locally if there is a problem with the connectivity (attempt to activate another profile and connect to another operator). In addition, it is scaling together with the IoT devices. Since this solution is completely under the control of the operator, it can be independent of the application, thereby also saving development costs.

Figure 3 shows an example of this system: a central ZTP service, in connection with multiple subsystems and a support application on the UICC module.

The central ZTP service working together with the ZTP support application on the UICC module can be very effective. The ZTP service and the ZTP support application together can cover almost

every use case and solve the problems the IoT area is struggling with today.

The UICC application can be used to monitor connectivity and fix issues locally. This can be highly effective if credentials are stored on the UICC module and if the IoT protocol stack is also running on the UICC module.

For narrowband IoT, the traditional profile download solution and the machine-to-machine SM-DP is ineffective. Significantly better results can be achieved by using the SM-DP+ in a new way. For example, running the LPA proxy on the UICC module makes it possible to use completely new options for device provisioning.

Conclusion

The universal integrated circuit card (UICC) modules present in all 3GPP IoT devices today are costly and underutilized.

The industry is looking for ways to replace them with a next-generation solution, but for the foreseeable future UICC modules are here to stay. While there are a few ways to reduce the complexity of using UICC modules and thereby reducing the cost of IoT devices, it is also possible to extend the application of UICC modules well

beyond the cellular domain. For example, members of the existing UICC ecosystem can start exploiting UICC capabilities for storing IoT identities, executing IoT protocols, increasing security, providing end-to-end connectivity as a service, and/or supporting zero-touch provisioning.

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Further reading

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FUTURE NETWORK TRENDS

CREATING INTELLIGENT DIGITAL INFRASTRUCTURE

BY: ERIK KUDDEN, CTO

All around the world, the unprecedented events of 2020 have brought into focus the critical role that digital infrastructure plays in the functioning of virtually every aspect of contemporary society. More than ever before, communication technologies are providing innovative solutions to help address social, environmental and economic challenges by enhancing efficiency and enabling both intensified network usage and more well-informed decisions.

One of the most important features of digital infrastructure is the ability to bridge distances and make it easier to efficiently meet societal needs in terms of resource utilization, collaboration, competence transfer, status verification, privacy protection, security and safety. The communications industry supports other industries by enabling them to deliver digital products and services such as health care, education, finance, commerce, governance and agriculture. It also plays a vital role in tackling climate change by helping other industries reduce emissions and improve efficiency.

In last year's trends article, I introduced the concept of the network platform and explained how it serves as a catalyst in the development of an open marketplace

that is always available to any consumer of the digital infrastructure. The network platform forms the core of the digital infrastructure, with the ability to ensure long-term competitiveness for enterprises and meet the full range of societal needs as well. It is a trustworthy solution that guarantees resilience, privacy, reliability and safety for all types of organizations – public, private and everything in between. It also has the scale, cost performance and quality required to support future innovations. As a result of these characteristics, it is the most sustainable solution to address all future communication needs.

Future technologies will enable a fully digitalized, automated and programmable world of connected humans, machines, things and places. All experiences and sensations will be transparent across the boundaries of physical and virtual realities. Traffic in future networks will be generated not only by human communication but also by connected, intelligent machines and bots that are embedded with artificial intelligence (AI). As time goes on, the percentage of traffic generated by humans will drop as that of traffic generated by machines and computer vision systems – including autonomous vehicles, drones and surveillance systems – rises.

The machines and other 'things' that make up the Internet of Things (IoT) require even more sophisticated communication than humans do. For example, connected, intelligent machines must be able to interact dynamically with the network. Sensor data will be used to support the development of pervasive cyber-physical systems consisting of physical objects connected to collaborative digital twins. Future network capabilities will also include support for the transfer of sensing modalities such as sensations and smell.

The network platform acts as a seamless universal connectivity fabric characterized by its almost limitless scalability and affordability. It is capable of exposing capabilities beyond communication services, such as embedded compute and storage as well as a distributed intelligence that supports users with insights and reasoning.

In this article, I will explain the ongoing evolution of the network platform in terms of the key needs that are driving its evolution (trends 1-3) and the emerging capabilities that will meet both those and other needs (trends 4-7).

Trends 1-3: The key drivers of network platform evolution

The three key drivers that are most significant to the evolution of the network platform are all related to bridging the gap between physical reality and the digital realm. Most notably, this involves delivering sensory experiences over networks and utilizing digital representations to make the physical world fully programmable.

TREND #1: A COLLABORATIVE, AUTOMATED PHYSICAL WORLD

As physical and digital realities become increasingly interconnected, advanced cyber-physical systems have begun to emerge. These systems consist of humans, physical objects (machines and other things), processes, networking and computation, and the interactions between them all. Their primary purpose is to provide individuals, organizations and enterprises with full transparency to monitor and control assets and places, thereby generating massive benefits in terms of efficiency. One early example of this is the way that cyber-physical systems can help planners optimize energy and materials usage.

Soon, there will be hundreds of billions of connected physical objects with embedded sensing, actuation and computing capabilities, which continuously generate informative data. The sensor data generated

by physical objects can be used to create their digital twins. Collaborative digital twins will have the ability to manage the interactions between the physical objects they represent.

Digitalizing the physical environment in which the physical objects interact requires sensor data fusion – that is, using data from multiple sources to create an accurate digital representation of the physical environment. One example of sensor data fusion is achieving high-precision positioning by combining network-based positioning data with information from other sensors such as cameras and inertial measurement units.

Ultimately, the joint communication and sensing in future systems will make it possible to leverage all the interconnected digital twins and digital representations of the environment to create a complete digital representation of everything.

TREND #2: CONNECTED, INTELLIGENT MACHINES

Machines will become increasingly intelligent and autonomous as their cognitive abilities continue to expand. Their understanding of the world around them will continue to grow in tandem with their ability to interact with other machines as part of a cognitive system of systems.

An intelligent machine uses sensors to monitor the environment and adjust its actions to accomplish specific tasks in the face of uncertainty and variability. These machines include three major subsystems: sensors, actuators and control. Examples of intelligent machines include industrial robots, speech recognition/voice synthesis and self-guided vehicles. The complexity of control and logic skills makes expert systems central in the realm of intelligent machines.

The network platform will provide an automated environment in which interconnected, intelligent machines can communicate, including support for AI-to-AI communication and autonomous systems such as communication among self-driving vehicles and intelligent machines in factories.

Intelligent machines have their own way of perceiving information (data), which is different from how humans perceive it. For example, communication among intelligent machines requires new types of video compression mechanisms, as today's video codecs are optimized for human perception.

Another aspect to consider is how intelligent machines will interact and communicate with each other. To improve the reliability and efficiency of machine-to-machine communication, machines will need to understand the meaning of the communication in terms of capabilities, intentions and needs. This will require semantics-driven communication.

Cognition is one of the most important capabilities of an intelligent machine. Cognitive machines are capable of self-learning from their interactions and experiences with their environment. They generate hypotheses and reasoned arguments, make recommendations and take actions. They can adapt and make sense of complexity and handle unpredictability. The future network will empower cognitive machines by providing them with new network features and services such as sensing, high-precision positioning and distributed computing capabilities.

TREND #3: THE INTERNET OF SENSES

The ability to deliver multisensory experiences over future networks will make it easier than ever before to transfer skills over the internet. It will ultimately lead to the emergence of

the internet of senses, which combines visual, audio, haptic and other technologies to allow human beings to have remote sensory experiences.

The internet of senses will enable seamless interaction with remote things and machines, making it possible to fully realize use cases such as remote health checks, remote operation of machinery, holographic communication and virtual reality (VR) vacations. Among other benefits, the internet of senses is expected to have a significant impact in terms of sustainability, by dramatically reducing the need for travel.

In the years ahead, major leaps forward are expected in sensor and actuator technologies, such as the actuation of smell and high-quality touch sensation. During remote operations, the advances in haptic devices will allow virtual objects to be perceived just as the real objects themselves. Holographic communication will be possible without wearing extended reality glasses, due to 3D light field display technologies.

Body augmentation capabilities will enable humans to be smarter, stronger and more capable. Other examples are contact lenses that can display augmented reality (AR) content, universal translator earbuds that allow for language-independent communication and exoskeletons that increase physical strength. Eventually, brain-computer interfaces will enable communication at the speed of thought where, instead of speaking to machines, humans will merely think in order to direct them.

The network platform supports the internet of senses with novel network enablers such as distributed compute, high-precision positioning, integrated sensing and application programming interfaces. These are needed to support bandwidth and latency reservation, network latency reporting and network slice prioritization.

USE CASE DIGITAL TWIN IN THE PORT OF LIVORNO

Ericsson has deployed a digital twin in the Italian port of Livorno (Leghorn). As a result, terminal port operations will increasingly become a mixture of physical machinery, robotics systems, automated vehicles, human-operated digital platforms and AI-based software systems. All those components, served by a 5G solution, transform the port environment into a 'playground' in which to experience the future of an automated physical world.

The port's digital twin makes use of a plethora of real-time data captured by connected objects at the physical port, including sensors, cameras and vehicles. An AI operation management system operates on the digital model to determine the sequence of logistics tasks and activities. Feedback from these processes provides live updates to the human supervisors using VR and to the docks/quay operators through AR.

Results indicate that there are about 60 direct and indirect benefits of the solution, including improved competitiveness, increased safety for personnel, sustainable growth of the port city, improved management of logistics and a positive environmental impact.

Trends 4-7: Critical enablers of the future network platform

The network platform is designed to deliver the kind of extreme performance required by application areas such as the internet of senses and communication among intelligent machines. It will also serve new types of devices with close-to-zero-cost and close-to-zero-energy implementations, which can be embedded into everything. Looking ahead, increasingly advanced technologies in four areas (trends 4-7) will expand the capabilities of the digital infrastructure through the network platform.

TREND #4:
OMNIPRESENT AND NON-LIMITING CONNECTIVITY

The concept of ubiquitous radio access is evolving toward the vision of a future network that will deliver non-limiting performance to satisfy the needs of humans, things and machines by enhancing multidimensional coverage, stellar capacity and augmenting capabilities.

Access coverage everywhere

Further densification of networks is needed to provide high-speed coverage everywhere. Connected airborne devices, such as drones, require access on altitudes up to several kilometers, making it necessary to have a 3D point of view including the elevation aspect to provide coverage. There is also a need to ensure high-performing indoor connectivity by increasing the number of indoor small cells and fully integrating them.

Flexible network topologies and deployments

Network topologies and deployments will need to become increasingly flexible to provide coverage everywhere and deliver extreme performance. One possibility is a multi-hop-based radio network, where a multitude of nodes collaborate to forward a message to the receiver. This solution is particularly interesting for smaller cells of limited reach. Satellites, high-altitude platforms and airborne cells can be integrated into the network as a complement to extend coverage. Further components in a flexible topology can include connected device relay and the possibility for ad-hoc deployments of networks. Ultimately, distributed massive MIMO (multiple-input, multiple-output) solutions may lead to fully distributed connectivity, where many radio network nodes simultaneously serve a user, without fixed-cell borders.

Access for zero-energy devices

The rapidly growing demand for vast numbers of connected sensors and actuators has made it necessary to invent zero-energy devices. These will be deployed once and will continuously report and act without the need for maintenance or external charging. The stepping stones along the way include narrowband IoT enhancements and massive machine type communication for 5G New Radio for local area networks (LANs) as well as for wide-area usage.

Extreme radio performance

The network will utilize higher frequency bands to deliver extreme performance. For example, communications over the terahertz frequency band (above 100GHz) have some attractive properties, including terabit-per-second link capacities and miniature transceivers.

The design of terahertz electronics includes very small antenna and radio frequency (RF) elements as well as high-performance oscillators.

Full duplex is another component that can, in some specific scenarios, substantially increase the link capacity compared with half duplex. Full duplex is made possible by self-interference suppression circuits.

Visible light wireless communication, piggybacking on the wide adoption of LED (light-emitting diode) lighting, is another potential step in the frequency domain to complement RF communications.

Network as a sensor

Higher frequencies will further enhance the spatial and temporal resolution of the radio signal. Reflections of such radio signals can be used to sense the surroundings. Furthermore, high frequencies have distinct atmospheric and material interactions, where different frequencies are more or less susceptible to things like absorption in water, for example. This has been shown to be sufficient to forecast weather and air quality.

Distance information to reflecting surfaces can be identified by integrating positioning and sensing capabilities. Such information can be used to detect obstacles and speed as well as to generate real-time local maps.

TREND #5:
PERVASIVE NETWORK COMPUTE FABRIC

As distributed compute and storage continues to evolve, the lines between the device, the edge of the network and the cloud will become increasingly blurred. Everything can be viewed as a single, unified, integrated execution environment for distributed applications, including both network functions and third-party applications. In the network compute

fabric, connectivity, compute and storage will be integrated, interacting to provide maximum performance, reliability, low jitter and millisecond latencies for the applications they serve.

Rather than processing data centrally, in many cases it is more efficient in terms of bandwidth and/or latency constraints to bring the processing closer to where the data is produced, insights are consumed and actions are taken. In some cases, local operation may be required by regulations or preferred for privacy, security or resilience reasons.

Aside from the applications, the network also provides a continuous execution environment for access and core functions. It runs on a distributed cloud infrastructure with integrated acceleration for data-intensive virtual network functions and applications.

The future network platform goes beyond the use of microservices to implement network functions as serverless architectures. The server management and capacity planning decisions are fully autonomous from the developer and the network operator. The network takes care



of the deployment, scaling and all resources required to ensure that the function deployed is always available at any scale.

Upcoming novel computing architectures include memory-centric computing, optical computing, nanocomputing, neuromorphic computing and even quantum computing. In the future, these architectures will enable continued exponential growth in compute capacity for most applications running on the network compute fabric – an important development as the end of Moore's law approaches.

TREND #6: TRUSTWORTHY INFRASTRUCTURE

Governments and enterprises are adopting advanced technologies for secure assurance of mission- and business-critical processes such as factory automation, remote control of assets and more. The highly trustworthy network platform fulfills the requirements of even the most mission- and business-critical use cases. It offers a fusion of connectivity and compute characterized by different dimensions of resilience, privacy, security, reliability and safety. It will also provide adaptable and verifiable dimensions of trustworthiness in a scalable and cost-efficient manner.

Rather than being designed per node or for a particular part of the network, the always-on characteristics of the network platform such as reliability, availability and resilience rise up to cover the complete network. Always-on mechanisms are built into user plane, control plane and device mobility solutions. All parts of the network will be addressed including transport nodes and transport networks, network infrastructure and site solutions.

To protect communication and data, secure identities are utilized at every layer between humans, devices and applications in different industry segments.

These identities are securely anchored to devices and network nodes by root-of-trust mechanisms.

Network platform solutions utilize confidential computing to protect identities and their data and establish trust among network customers and their assets, thereby also offering assurance to users and regulators. This requires automated trust assessment of all network elements, things, machines and applications, as well as compute and storage resources by using remote attestation and AI.

Responsible AI will bring trustworthy automated protection and risk management. AI-based automation provides the ability to act on a high number of events affecting the network infrastructure or the network usage.

TREND #7: COGNITIVE NETWORK

In the vision of zero-touch network management and operations, networks are deployed and operated with minimum human intervention, using trustworthy AI technologies. All operational processes and tasks, including, for example, delivery, deployment, configuration, assurance and optimization, will be executed with 100 percent automation.

The network itself will continuously learn from its environment observations, interactions with humans and previous experiences. The cognitive processes understand the current network situation, plan for wanted outcome, decide on what to do and act accordingly. The outcome serves as an input to learn from its actions. The cognitive network will be able to optimize its existing knowledge, build on experience and reason in order to solve new problems.

The network will utilize intent-based and distributed intelligence for multiple functions, including optimization of the

radio interface, automation of network management and orchestration such as the optimization of parameters, handling of alarms and self-healing. AI algorithms will be deployed and trained at different network domains, for example, in management, the core network and the radio network. Physical layer algorithms, such as link adaptation, handover, power control and dynamic scheduling of resources can be optimized with AI agents.

Network management will become less complex through intelligent closed-loop automation with support for humans to interact with the network and monitor its behaviors. The network operator expresses the intent of a desired network state or goal, and the network internally resolves the detailed steps necessary to achieve that intent. Network knowledge, data and actions are shaped in such a way that the operator interacting with the network can understand them.

The cognitive network will be based on control design, using both machine reasoning and machine learning techniques that are distributed and capable of acting in real time. The network is a highly distributed system where multiple AI agents, present across the network, need to interwork to optimize overall network performance. Local decisions need to be coordinated with more central intent-based decisions. The central AI agent needs to make decisions in real time based on both local and global information. Multiple distributed AI agents share distributed insights throughout the network through federated learning.

Cognitive networks will be inherently trustworthy – that is, reliable, safe, secure, fair, transparent, sustainable and resilient – by design.



CONCLUSION

The network platform is the spinal cord of intelligent digital infrastructure

The digital infrastructure offers endless possibilities to individuals, enterprises and governments across the globe, with its unique ability to bridge vast distances and enable powerful new solutions to a wide range of social, environmental and economic challenges. Health care, education, finance, commerce, governance and agriculture are just a few of the sectors that stand to benefit from the massive efficiency gains that digital infrastructure can provide.

Designed to carry vital messages, commands, reasoning, insights, intelligence and all the sensory information needed to support the continuous evolution of industry and society, the network platform is designed to be the spinal cord of digital infrastructure.

It is also the ideal platform for all types of innovation, with the ability to support interactions that empower an intelligent, sustainable and connected world. The major advantage of the network platform is that it will be accessible anywhere, always-on and with guaranteed performance. Nomadic distributed processing and storage will be embedded into it to support advanced applications. It will be inherently reliable and resilient, fulfilling all the requirements for secure communication. Cognitive operations and maintenance of the network and its services will deliver the most cost-efficient and sustainable solution to meet any and all communication needs.

With this in mind, it is clear that the most important future network trends to watch in 2020 are those that relate most closely to

the growth and expansion of intelligent digital infrastructure on the network platform. The first three of the seven trends this year are the key drivers of network platform evolution – the creation of a collaborative automated physical world, connected, intelligent machines and the internet of senses. All three highlight the growing need to bridge the gap between physical and digital realities. Trends 4-7 are increasingly advanced technologies in four areas – non-limiting connectivity, pervasive network compute fabric, trustworthy infrastructure and cognitive networks. Breakthroughs in these four areas will be essential to fully enable trends 1-3 and continuously expand the capabilities of the digital infrastructure through the network platform in the years and decades ahead.



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The future of cloud computing

HIGHLY DISTRIBUTED WITH HETEROGENEOUS HARDWARE

With a vastly distributed system (the telco network) already in place, the telecom industry has a significant advantage in the transition toward distributed cloud computing. To deliver best-in-class application performance, however, operators must also have the ability to fully leverage heterogeneous compute and storage capabilities.

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The cloud is transforming, both in terms of the extent of distribution and in the diversity of compute and storage capabilities. On-premises and edge data centers (DCs) are emerging, and hardware (HW) accelerators are becoming integral components of formerly software-only services.

■ One of the main drivers into the age of virtualization and cloud was the promise of reducing costs by running all types of workloads on homogeneous, generic, commercial off-the-shelf (COTS) HW hosted in dedicated, centralized DCs. Over the years, however, as use

cases have matured and new ones have continued to emerge, requirements on latency, energy efficiency, privacy and resiliency have become more stringent, while demand for massive data storage has increased.

To meet performance, cost and/or legal requirements, cloud resources are moving toward the edge of the network to bridge the gap between resource-constrained devices and distant but powerful cloud DCs. Meanwhile, traditional COTS HW is being augmented by specialized programmable HW resources to satisfy the strict performance requirements of certain applications and limited energy budgets of remote sites.

The result is that cloud computing resources are becoming increasingly heterogeneous, while simultaneously being widely distributed across smaller DCs at multiple locations. Cloud deployments must be rethought to address the complexity and technical challenges that result from this profound transformation.

In the context of telecommunication networks, the key challenges are in the following areas:

- 1. Virtualization of specialized HW resources
- 2. Exposure of heterogeneous HW capabilities
- 3. HW-aware workload placement
- 4. Managing increased complexity.

Getting all these pieces right will enable the future network platform to deliver optimal application performance by leveraging emerging HW innovation that is intelligently distributed throughout the network, while continuing to harvest the operational and business benefits of cloud computing models.

Figure 1 positions the four key challenges in relation to the orchestration/operations support systems (OSS) layer, the application layer, run-time and the operating system/hypervisor. The lower part of the figure provides some examples of specialized HW in a telco environment, which includes domain-specific accelerators, next-generation memory and storage, and novel interconnect technologies.

Compute and storage trends

With the inevitable end of Moore's Law [2], developers can no longer assume that rapidly increasing application resource demands will be addressed by the next generation of faster general-purpose chips. Instead, commodity HW is being augmented by a very heterogeneous set of specialized chipsets, referred to as domain-specific accelerators, that attempt to provide both cost and energy savings.

For instance, data-intensive applications can take advantage of the massive scope for parallelization

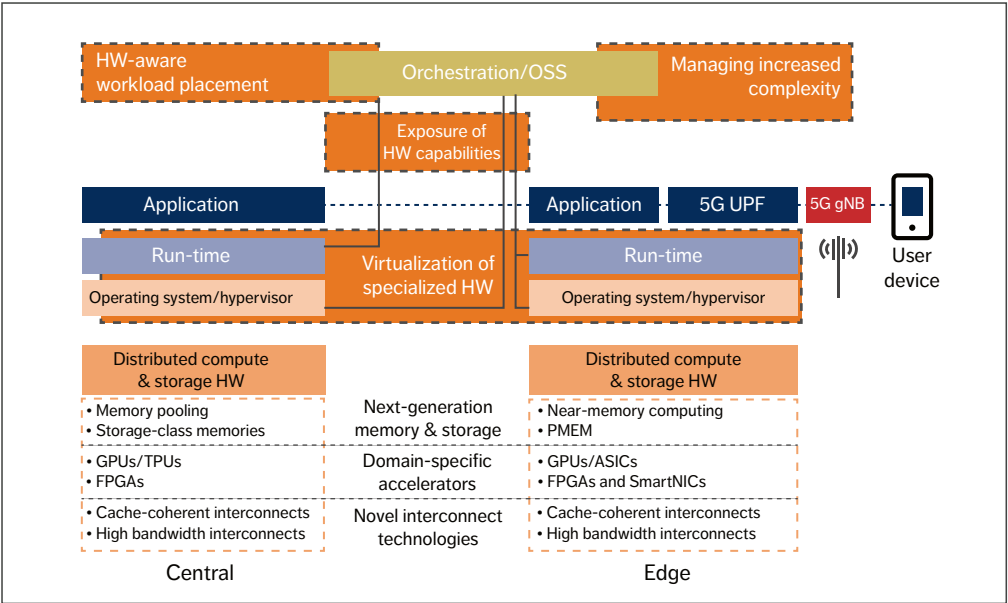


Figure 1 Impact of the four key challenges on the stack (top) and heterogeneity of HW infrastructure (bottom)

in graphics processing units (GPUs) or tensor processing units (TPUs), while latency-sensitive applications or locations with limited power budgets may utilize field-programmable gate arrays (FPGAs). These trends point to a rapidly increasing adoption of accelerators in the near future.

The growing demand for memory capacity from emerging data-intensive applications must be met by upcoming generations of memory. Next-generation memories aim to blur the strict dichotomy between classical volatile and persistent storage technologies – offering the capacity and persistence features of storage, combined with the byte-addressability and access speeds close to today’s random-access memory (RAM) technologies. Such persistent memory (PMEM) technologies [3] can be used either as large terabyte scale volatile memory, or as storage with better latency and bandwidth relative to solid-state disks.

●● THESE TRENDS POINT TO A RAPIDLY INCREASING ADOPTION OF ACCELERATORS... ●●

3D silicon die-stacking has facilitated the embedding of compute units directly inside memory and storage fabrics, opening a paradigm of near-memory processing [1], a technology that reduces data transfer between compute and storage and improves performance and energy consumption.

Finally, advancements in interconnect technologies will enable faster speeds, higher capacity and lower latency/jitter to support communication between the various memory and processing resources within nodes as well as within clusters. The cache coherency properties of modern interconnect technologies, such as Compute Express Link [4] and Gen-Z, can enable direct access to configuration registers and memory regions across the compute infrastructure. This will simplify the programmability of accelerators and facilitate fine-grained data sharing among heterogeneous HW.

Supporting heterogeneous hardware in distributed cloud

While the combination of heterogeneous HW and distributed compute resources poses unique challenges, there are mechanisms to address each of them.

Virtualization of specialized hardware

The adoption of specialized HW in the cloud enables multiple tenants to use the same HW under the illusion that they are the sole user, with no data leakage between them. The tenants can request, utilize and release accelerators at any time using application programming interfaces (APIs). This arrangement requires an abstraction layer that provides a mechanism to schedule jobs to the specialized HW, monitor their resource usage and dynamically scale resource allocations to meet performance requirements. It is pertinent to keep the overhead of this virtualization to a minimum. While virtualization techniques for common COTS HW (x86-based central processing units (CPUs), dynamic RAM (DRAM), block storage and so on) have matured well during recent decades, corresponding virtualization techniques for domain-specific accelerators are largely still missing for production-grade systems.

Exposure of hardware capabilities

Current cloud architectures are largely agnostic to the capabilities of specialized HW. For example, all GPUs of a certain vendor are treated as equivalent, regardless of their exact type or make. To differentiate them, operators typically tag the nodes equipped with different accelerators with unique tags and the users request resources with a specific tag. This model is very different to general-purpose CPUs and can therefore lead to complications when a user requires combinations of accelerators.

Current deployment specifications also do not have good support for requesting partial allocation of accelerators. For accelerators that can be partitioned today, the decomposition of a single

physical accelerator into multiple virtual accelerators must be done manually. Addressing these issues will require appropriate abstractions and models of specialized HW, so that their capabilities can be interpreted and incorporated by orchestration functions.

The need for appropriate models will be further amplified in the case of distributed compute and storage. Here, the selection of the optimal site location will depend on the application requirements (bounded latency or throughput constraints, for example) and the available resources and HW capabilities at the sites. The programming and orchestration models must be able to select appropriate software (SW) options – SW only in the case of moderate requirements, for example, or SW complemented with HW acceleration for stringent requirements.

As SW deployment options with or without HW acceleration may have significantly different resource footprints, sites must expose their HW capabilities to be able to construct a topology map

of resources and capabilities. During exposure and abstraction, proprietary features and the interfaces to them must be hidden and mapped to (formal or informal) industry standards that are hopefully coming soon. Modeling and abstraction of resources and capabilities are necessary prerequisites to be able to select the appropriate location and application deployment options and flavors.

Orchestrating heterogeneous distributed cloud

Based on a global view of the resources and capabilities within the distributed environment, an orchestration system (OSS in telco terminology) typically takes care of designing and assigning application workloads within the compute and storage of the distributed environment. This means that decisions regarding optimal workload placement also should factor in the type of HW components available at the sites related to the requirements of the specific application SW.

Due to the pricing of and power constraints on existing and upcoming HW accelerators,

Definition of key terms

Edge computing provides distributed computing and storage resources closer to the location where they are needed/consumed.

Distributed cloud provides an execution environment for cloud application optimization across multiple sites, including required connectivity in between, managed as one solution and perceived as such by the applications.

Hardware accelerators are devices that provide several orders of magnitude more efficiency/performance compared with software running on general purpose central processing units for selected functions. Different hardware accelerators may be needed for acceleration of different functions.

Persistent memory is an emerging memory technology offering capacity and persistence features of block-addressable storage, combined with the byte-addressability and access speeds close to today’s random-access memory technologies. It is also referred to as storage-class memory.

Moore’s law holds that the number of transistors in a densely integrated circuit doubles about every two years, increasing the computational performance of applications without the need for software redesign. Since 2010, however, physical constraints have made the reduction in transistor size increasingly difficult and expensive.

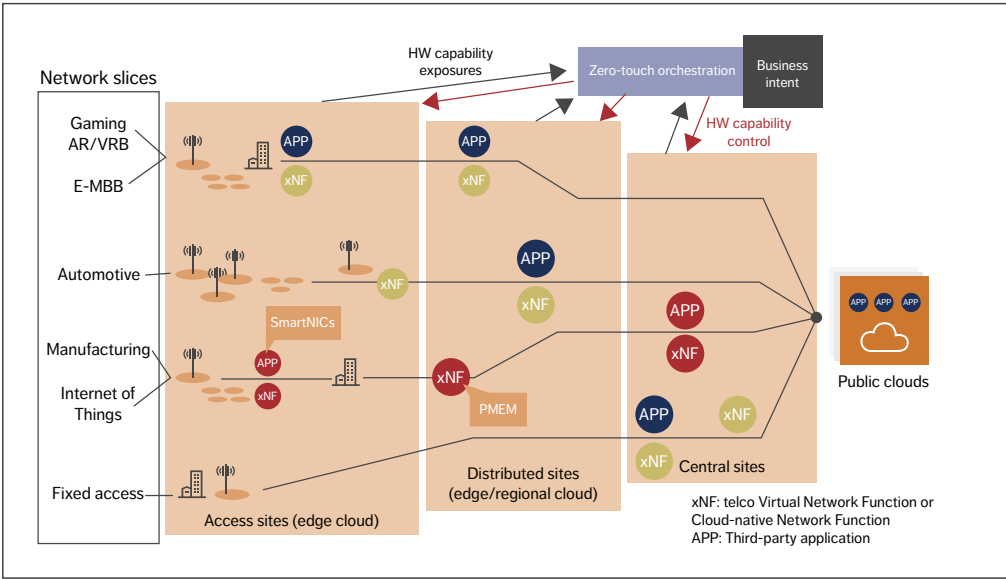


Figure 2 Integrated network slicing (telco) and third-party applications

they are expected to be scarce among edge-cloud sites, which in turn will require mechanisms to employ prioritization and preemption of workloads. Unlike conventional IT cloud environments, distributed cloud allows considerations of remote resources and capabilities.

Moreover, telco applications and workloads hosted in telco clouds may require much stricter Service Level Agreements (SLAs) to be fulfilled. Prioritization and preemption for new workloads may only be a viable option if capabilities or resources are already taken. However, it is important to migrate evicted workloads either to a new location, or to a new SW and HW deployment option to minimize service disruption during preemption.

Managing increased complexity

Traditional automation techniques based on human scripting and/or rule books cannot scale to address the complexity of the heterogeneous distributed cloud. We can already see a shift away

from human-guided automation to machine-reasoning-based automation such as cognitive artificial intelligence (AI) technologies. Specifically, a paradigm is emerging where the human input to the cloud system will be limited to specifying the desired business objectives (intents). The cloud system then figures how best to realize those objectives/intents.

Figure 2 presents an exemplary distributed cloud scenario with access sites, regional and central DCs and public clouds. It is based on the assumption that the manufacturing network slice (red) includes both telco (xNF) and third-party workloads (APP), out of which one APP requires network acceleration (SmartNIC), while another xNF depends on PMEM.

Multiple network slices are created based on customer need. Network slices differ not only in their service characteristics, but are separated and isolated from each other. Aggregated views of HW accelerators per location are collected for the zero-touch orchestration service (gray arrows).

When a service request arrives, the orchestration service designs the service instance topology and assigns resources to each service component instance (red arrows). These actions are based on the actual service requirements, the service access points and the business intent.

Opportunities and use cases

In terms of the opportunities in support of the ongoing cloudification of telco networks, let us consider the case of RAN. The functional split of higher and lower layers of the RAN protocol makes it possible to utilize Network Functions Virtualization (NFV) and distributed compute infrastructure to achieve ease of deployment and management. The asynchronous functions in the higher layer may be able to be run on COTS HW.

However, a set of specialized HW will be required to meet the stringent performance criteria of lower-layer RAN functions. For instance, the time-synchronous functions in the medium-access control layer, such as scheduling, link adaptation, power control, or interference coordination, typically require high data rates on their interfaces that scale with the traffic, signal bandwidth and number of antennas. These cannot be easily met with current general-purpose processing capabilities.

Likewise, deciphering functions in the packet data convergence protocol layer, compression/decompression schemes of fronthaul links and channel decoding and modulation functions in the physical layer would all benefit from HW acceleration.

The security requirements for data flows across the backhaul for 4G/5G RANs mandate the use of IP security protocols (IPsec). By offloading encrypt/decrypt functions to specialized HW such as Smart Network Interface Controllers (SmartNICs), application-specific integrated circuits (ASICs) or FPGAs, the processing overhead associated with IPsec can be minimized. This is crucial to support higher data rates in the transport network.

The network data analytics function in 5G Core networks would benefit from GPUs to accelerate

training of machine learning (ML) models on live network data. The enhancements to interconnects (cache coherency, for example) make it easier for the various accelerators and CPUs to work together. The interconnects also enable low latencies and high bandwidths within sites and nodes. There is increasing demand on memory from several core network functions (user-database functions, for example), both from a scale and a latency perspective. The scale of PMEM can be intelligently combined with the low latency of double data rate memories to address these requirements.

While these opportunities are specific to telecommunication providers, there are also several classes of third-party applications that would benefit from distributed compute and storage capabilities within the telco infrastructure. Industry 4.0 includes several use cases that could utilize HW-optimized processing. Indoor positioning typically requires the processing of high-resolution images to accurately determine the location of an object relative to others on a factory floor. This is computationally intensive and GPUs/FPGAs are typically used. Likewise, the application of augmented reality (AR)/virtual reality (VR) technologies in smart manufacturing for remote assistance, training or maintenance will rely significantly on HW acceleration and edge computing to optimize performance and reduce latencies.

The gaming industry is also witnessing significant technology shifts – specifically, remote rendering and mixed-reality technologies will have a profound impact on the consumer experience. These technologies rely on an underlying distributed cloud infrastructure that has HW acceleration capabilities at the edge to offload the processing from consumer devices, while maintaining strict latency bounds.

Furthermore, several use cases in the automotive industry involve strict latency requirements that demand HW acceleration in the form of GPUs and FPGAs at remote sites. Examples include real-time object detection in video streams that are processed by either vehicles or road-side infrastructure.

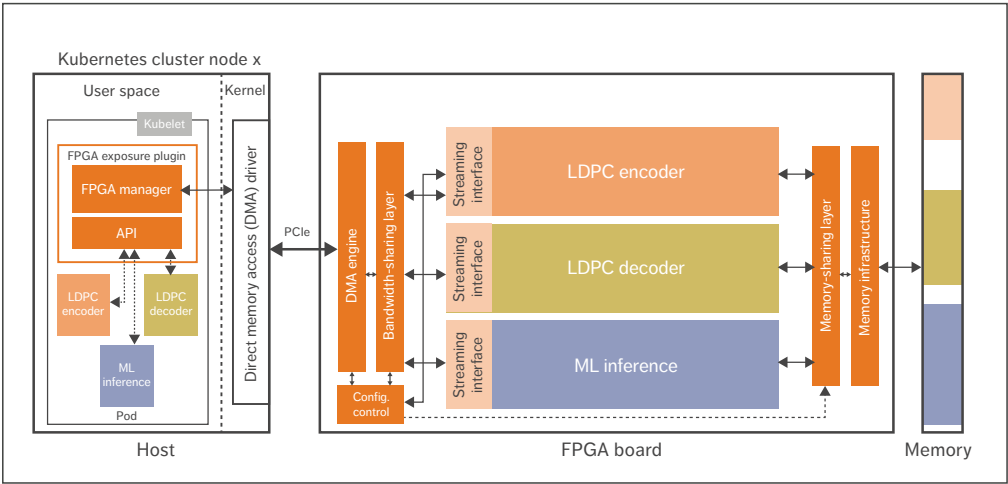


Figure 3 FPGA sharing solution

Initial proof points

Our ongoing research in the area of distributed cloud has yielded several initial proof points that demonstrate Ericsson’s leadership in terms of fully leveraging heterogeneous compute and storage capabilities to deliver best-in-class application performance to our customers.

Virtualized heterogeneous hardware

The 3GPP network evolution requires a dramatic increase in compute capacity when increasing carrier bandwidths up to terahertz level and the addition of more MIMO (multiple-input, multiple-output) layers and antenna ports. High-end GPUs could provide the required compute capacity. To understand how 5G baseband can be implemented on GPUs, Ericsson has entered into a collaboration with NVIDIA. Early findings show that the GPU instruction set and CUDA (Compute Unified Device Architecture) SW design patterns for key receiver algorithms excel in comparison with CPU-only solutions, indicating that further investigation of GPUs for this purpose is indeed a viable direction.

As high-end GPUs tend to be expensive and may not be available in every node of the distributed cloud, Ericsson Research has also devised a solution to enable GPU access by applications hosted on nodes without local GPUs, both enabled by OpenStack and Kubernetes. Given a high-speed network between the nodes, GPU requests are locally intercepted and redirected to the remote GPU servers for execution.

FPGAs may be another alternative to meet the requirements of RAN functions and third-party applications hosted at the telco edge. At Ericsson Research, we have enabled multi-tenancy support on FPGAs through Kubernetes, so that multiple-application containers that need HW acceleration can share the FPGA’s internal resources, off-chip DRAM and peripheral component interconnect express (PCIe) bandwidth, as shown in Figure 3.

On the left side, three application containers (pods) on a node are managed through Kubernetes. Our FPGA exposure plugin supports the pods by offering three isolated, virtual FPGAs. On the right side, three separate FPGA regions (one per application container) are managed and exposed

through our HW management layer, comprising of PCI bandwidth and memory-sharing functions (the orange boxes in the FPGA part of the figure).

An HW management layer uses partial reconfiguration technology to support spatial partitioning of one FPGA to share its resources. It is comprised of a bandwidth-sharing layer for dynamic allocation of PCIe bandwidth. The memory-sharing layer protects tenants from data leaks on the off-chip DRAM. It adds a protection layer for internal reconfiguration to prevent unintended configurations. This solution has been validated using multiple regions, hosting 5G low-density parity-check (LDPC) code encoder and decoder accelerators next to an ML inference function. However, essentially any telco or third-party application accelerator could be deployed within these partial regions, whose number could vary depending on the FPGA size and the application’s resource requirements.

Virtual switches form an integral part of distributed cloud infrastructure, providing network access to virtual machines by linking the virtual and physical network interfaces. The overhead of these virtual switches is one of the main obstacles to achieving high throughput in the packet-processing functions. Through our solution to off-load the data plane of a virtual switch onto the specialized HW

(FPGA-based SmartNICs in this case), we achieve the highest possible network input/output performance in a virtualized environment and free up CPUs for the execution of other functions and applications.

Orchestrating heterogeneous distributed cloud

There is a need for intelligent workload placement schemes to meet the diverse acceleration needs of 5G use cases that require domain-specific HW. We have two proof points related to this issue.

The first is our development of an HW-aware service instance design and workload placement to optimize the deployment of workloads at the edge. We have demonstrated an on-demand service instance design, prioritization and preemption for the telco Packet Core Gateway (PCG) user-plane function (UPF) over a Kubernetes edge cluster, in which only some nodes are equipped with PMEM. The PCG-UPF is considered a high-priority workload, which could use PMEM to distribute database instances needed for resiliency.

At the instantiation time of the PCG-UPF, another workload using the PMEM was evicted. The challenge was to migrate the lower priority workload to a new host, considering its original service requirements. Here, the lower priority workload shared an affinity with other components,

Terms and abbreviations

AI – Artificial Intelligence | API – Application Programming Interface | AR – Augmented Reality | ASIC – Application-Specific Integrated Circuit | COTS – Commercial Off-the-Shelf | CPU – Central Processing Unit | DC – Data Center | DRAM – Dynamic Random-Access Memory | E-WBB – Enhanced Mobile Broadband | FPGA – Field-Programmable Gate Array | GNB – GNodeB (3GPP next-generation base station) | GPU – Graphics Processing Unit | HW – Hardware | IPsec – IP Security Protocols | LDPC – Low-Density Parity-Check | ML – Machine Learning | NFV – Network Functions Virtualization | OSS – Operations Support Systems | PCG – Packet Core Gateway | PCIe – Peripheral Component Interconnect Express | PMEM – Persistent Memory | RAM – Random-Access Memory | SLA – Service Level Agreement | SmartNIC – Smart Network Interface Controller | TPU – Tensor Processing Unit | UPF – User-Plane Function | VR – Virtual Reality | xNF – network functions, both telco and third party

which were automatically migrated together with the workload blocking the PMEM needed by the PCG-UPF. The service disruption time for the evicted traffic was minimized with the instant triggering of the migration workflow.

Our second proof point demonstrates how we can enable distributed applications to utilize the enhanced capacity and persistency of non-volatile memories in a transparent fashion. While PMEM could be used to make up for the slow growth of DRAM capacity in recent years, it can lead to performance degradation due to PMEM's slightly higher latencies.

The memory-tiering concept developed by Ericsson Research enables the dynamic placement/migration of application data across DRAM and PMEM, based on observed application behavior. This infrastructure, using low-level CPU performance counters to drive placement decisions, achieves performance similar to DRAM, without any changes to the application, while using a mix of PMEM and DRAM.

... PAVING THE WAY TOWARD AN INTEGRATED NETWORK COMPUTE FABRIC THAT IS UNIVERSALLY AVAILABLE

Complexity management

A heterogeneous and distributed cloud implies high complexity in service assurance, and more specifically, high complexity in terms of continually finding optimal configurations in dynamically changing environments. Ericsson's cognitive layer has demonstrated cloud service assurance while satisfying contracted SLAs.

This cognitive process evaluates the effect of a proposed new service deployment on all existing services and their SLA fulfilment. Furthermore, the cognitive layer continuously reevaluates whether the current deployment of all services

is still optimal, and searches for improved configurations and uses predictive models to drive proactive actions to be taken, thereby enabling intelligent autonomous cloud operations.

Conclusion

The cloud is becoming more and more distributed at the same time that compute and storage capabilities are becoming increasingly diverse. Datacenters are emerging at the edges of telco networks and on customer premises and hardware accelerators are becoming essential components of formerly software-only services. Due to the inevitable end of Moore's law, the importance and use of hardware accelerators will only continue to increase, presenting a significant challenge to existing solutions for exposure and orchestration.

To address these challenges, Ericsson is innovating in three key areas. Firstly, we are using virtualization techniques for domain-specific accelerators to support sharing and multi-tenant use of specific hardware. Secondly, we are using zero-touch orchestration for hardware accelerators, which includes hardware capability exposure and aggregation for the orchestration system, as well as automated mechanisms to design and assign service instances based on the abstract map of resources and accelerator capabilities. And thirdly, we are using artificial intelligence and cognitive technologies to address the technical complexity and to optimize for business value.

Redefining cloud to expose and optimize the use of heterogeneous resources is not straightforward, and to some extent goes against the centralization and homogenization trends. However, we believe that our use cases and proof points validate our approach and will gain traction both in the telecommunications community and beyond, paving the way toward an integrated network compute fabric [5] that is universally available across telco networks.

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Critical IoT connectivity

IDEAL FOR TIME-CRITICAL COMMUNICATIONS

Critical Internet of Things (IoT) connectivity is ideal for a wide range of time-critical use cases across most industry verticals, and mobile network operators are uniquely positioned to deliver it.

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Cellular Internet of Things (IoT) is driving transformation across various sectors by enabling innovative services for consumers and enterprises. There are currently more than one billion cellular IoT connections, and Ericsson forecasts that there will be around five billion connections by 2025 [1].

■ As 5G deployments gain momentum globally, enterprises in almost every industry are exploring the potential of 5G to transform their products, services and businesses. Since the requirements for wireless connectivity in different industries vary, it is useful to group them into four distinct IoT connectivity segments: Massive IoT, Broadband IoT, Critical IoT and Industrial Automation IoT [2].

While Massive IoT and Broadband IoT already exist in 4G networks, Critical IoT will be introduced

with more advanced 5G networks. Industrial Automation IoT, the fourth segment, includes capabilities on top of Critical IoT that enable integration of the 5G system with real-time Ethernet and time-sensitive networking (TSN) used in wired industrial automation networks.

Critical IoT addresses the time-critical communication needs of individuals, enterprises and public institutions. It is intended for time-critical applications that demand data delivery within a specified time duration with required guarantee (reliability) levels, such as data delivery within 50ms with 99.9 percent likelihood (reliability).

Critical IoT is a paradigm shift from the enhanced mobile broadband (eMBB) connectivity, where the data rate is maximized without any guarantee on latency [3]. Many industry sectors have already started piloting time-critical use cases.

Time-critical use cases

The majority of time-critical use cases can be classified into the following four use case families:

- » Industrial control
- » Mobility automation
- » Remote control
- » Real-time media

Each family is relevant for multiple industries and includes a wide range of use cases with more or less stringent time-critical requirements, as shown in *Figure 1*.

Furthermore, there are three main network deployment scenarios depending on the coverage needs of time-critical services in different industries:

- » Local area
- » Confined wide area
- » General wide area

Local-area deployment includes both indoor and outdoor coverage for a small geographical area such as a port, farm, factory, mine or hospital. Confined wide-area deployment is for a predefined geographical area – along a highway, between certain electrical substations, or within a city center, for example. General wide-area deployment is about serving devices virtually anywhere.

Common to all time-critical use cases is the fact that the communication service requirements depend on the dynamics of the use case and the application implementation. A highly dynamic system requires faster control with shorter round-trip times (RTTs), while a slower control loop is sufficient for a system that operates more slowly.

Various factors – such as device processing capabilities, the processing split between the device and the application server, the application’s ability to extrapolate and predict data in case of missing

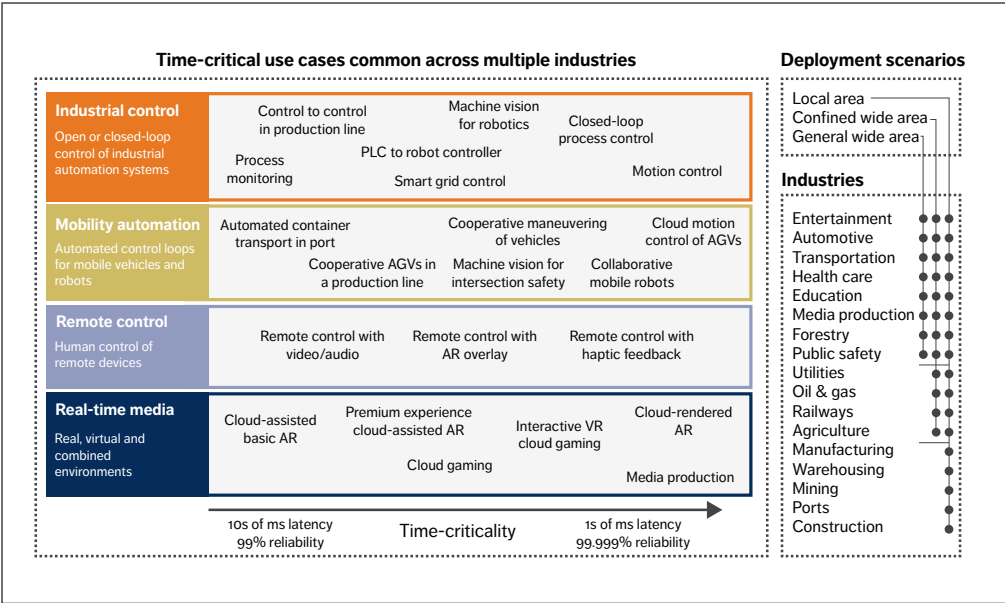


Figure 1 Examples of use cases enabled by Critical IoT

Time-critical use case trials

In partnership with leading industry partners and mobile network operators, Ericsson has trialed various Critical IoT use cases including:

- » **Industrial control for manufacturing vehicles:** <https://www.ericsson.com/en/networks/cases/accelerate-factory-automation-with-5g-urllc>
- » **Industrial control for manufacturing jet engines:** https://www.youtube.com/watch?v=XZWC_ttighM
- » **Remote control in mining:** <https://www.youtube.com/watch?v=C4lOUKZ-FCc&t=7s>
- » **Remote control of autonomous trucks:** <https://www.ericsson.com/en/press-releases/2018/11/ericsson-einride-and-telia-power-sustainable-self-driving-trucks-with-5g>
- » **Remote bus driving:** <https://www.youtube.com/watch?v=IPyzGTD5FtM>
- » **Cooperative vehicle maneuvers:** <https://5gcar.eu/>
- » **Virtual reality and real-time media:** <https://www.ericsson.com/en/blog/2017/5/its-all-green-flags-for-5g-at-the-indianapolis-motor-speedway>
- » **Augmented reality:** <https://www.ericsson.com/en/news/2018/3/5g-augmented-reality>
- » **Smart harbor:** <https://www.ericsson.com/en/press-releases/2019/2/ericsson-and-china-unicom-announce-5g-smart-harbor-at-the-port-of-qingdao>

packets, rate adaptivity and which codecs are used – impact both the application RTT and the latency requirements on the communication network.

Industrial control includes a very broad set of applications, present in most industry verticals [4]. These applications typically consider late messages as lost. Process monitoring, controller-to-controller communication between production cells and some control functions for the electricity grid are examples of use cases with modest time-criticality, while use cases such as closed-loop process control and motion control have very stringent requirements.

Mobility automation refers to the automation of control loops for mobile vehicles and robots. Examples of the least time-critical use cases in this category include the relatively self-sufficient automated guided vehicles (AGVs) equipped with advanced on-board sensors that are used for transportation in ports and mines. Infrastructure-assisted vehicles such as fast-moving AGVs

in a warehouse and collaborative maneuvering on public roads are examples of more time-critical mobility automation use cases, while the collaborative mobile robots used in flexible production cells represent an even higher degree of time-criticality.

Remote control refers to the remote control of equipment by humans. The ability to remotely control equipment is an important step in the evolution toward autonomous vehicles (to take temporary control of a driverless bus in scenarios not covered by its own automation functions) and for flying drones beyond visual line-of-sight.

Remote control can also improve work environments and productivity by moving humans out of inconvenient or hazardous environments – remote-controlled mining equipment [5] is one example. Such solutions also offer the benefit of providing enterprises with access to a broader workforce.

The communication service requirements for remote control depend on how fast the remote environment changes, the required precision of the task and the required QoE. Control-loop latency and audio/video quality are important factors for QoE and the ergonomics for the remote operator. Haptic feedback and augmented reality (AR) can be used to further improve the operator QoE and task precision, and will make the acceptable latencies even stricter.

Real-time media comprises use cases where media is produced and consumed in real time, and delays have a negative impact on QoE. Mobile applications for gaming and entertainment, including AR and virtual reality (VR), are common, with processing and rendering done locally in the device. Time-critical communication will make it possible to offload parts of the processing and rendering to the cloud [6], thereby improving the user experience and enabling the use of more lightweight devices (head-mounted, for example).

Time-critical communication can enable cloud gaming over cellular networks as well as new applications in sectors such as manufacturing, education, health care and public safety. It is expected to drive more widespread use of mobile AR and VR. Advanced media production (such as real-time production of live performances) with its strict delay and synchronization requirements, is another area where time-critical communication can enable new use cases.

Key network technologies and architectures

Achievable end-to-end (E2E) latencies depend on the available network and compute infrastructure, software features, and how the use case is implemented. In remote control, the physical distance between the remote operator and the teleoperated equipment is a physical property of the use case. In other use cases, the physical distance between end nodes can be reduced by distributed cloud processing, as in AR cloud gaming, where the AR overlay can be rendered in an edge cloud to limit interaction latencies. Network orchestration optimizes the placement of network and application functions to ensure efficient use of the compute and network infrastructure while restricting the transmission paths according to latency needs [7].

The 5G network comprises two functional domains: the next generation (5G) RAN (NG-RAN) and the 5G Core (5GC), which are built on an underlying transport network. All three – the NG-RAN, the 5GC and the transport network – contribute to the E2E reliability and latency, which is further affected by the device implementation.

The NG-RAN is deployed in a distributed fashion to provide radio coverage with good performance, availability and capacity. The 5GC provides connectivity of the device to the external services and applications. The network latency between the application and the RAN can be a major contributor to E2E latency.

Terms and abbreviations

- 5GC** – 5G Core | **AAS** – Advanced Antenna System | **AGV** – Automated Guided Vehicle | **AR** – Augmented Reality | **CA** – Carrier Aggregation | **DC** – Data Center | **DL** – Downlink | **E2E** – End-to-End | **eMBB** – Enhanced Mobile Broadband | **FDD** – Frequency Division Duplex | **IoT** – Internet of Things | **MNO** – Mobile Network Operator | **NG-RAN** – Next Generation RAN | **NPN** – Non-Public Network | **NR** – New Radio | **PLC** – Programmable Logic Controller | **RTT** – Round-Trip Time | **TDD** – Time Division Duplex | **TSN** – Time-Sensitive Networking | **UE** – User Equipment | **UL** – Uplink | **URLLC** – Ultra-Reliable Low-Latency Communication | **VR** – Virtual Reality

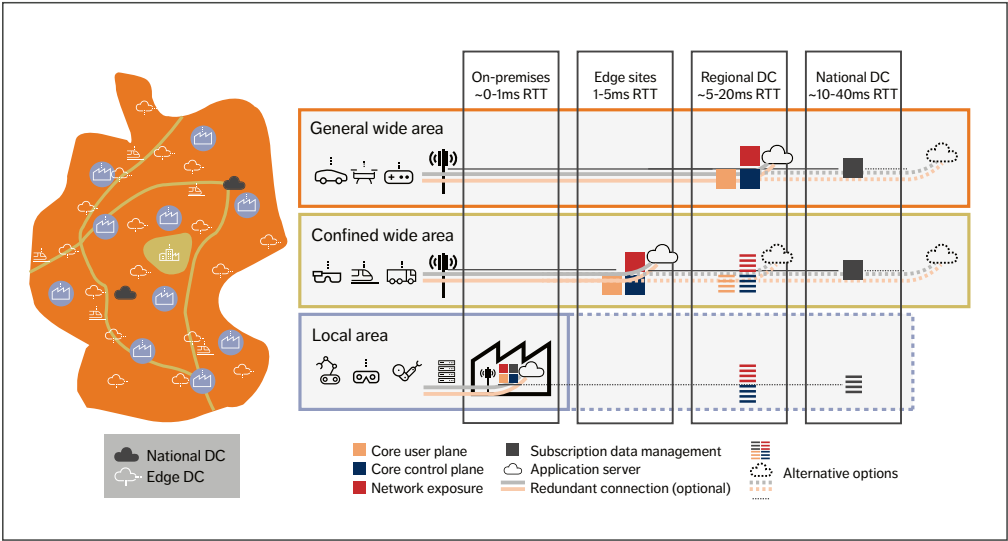


Figure 2 Examples of network architectures for low latency and/or high reliability

Figure 2 provides examples of network architectures for low latency and/or high reliability, and illustrates the effect of moving the application closer to the device. If an application is hosted in a central national data center (DC), the transport network round-trip latency can be in the order of 10-40ms, depending on the distance to the DC and how well the transport network is built out. Transport latency can be reduced to 5-20ms by moving applications to a regional DC or even to 1-5ms for edge sites. For local network deployments with networking functions and applications hosted on-premises, transport latencies become negligible.

Control of the network topology and the transport latency can be achieved by placing virtualized core network functions for execution at any location within the distributed computing platform of the network. This software-based design provides flexibility in updating the network with new functionality and reconfiguring it according to requirements. In addition to running

telecommunication functions, the distributed computing platform allows the hosting of application functions in the network [8].

Network slicing makes it possible to create multiple logical networks that share a common network infrastructure. A dedicated network slice can be created by configuring and connecting computing and networking resources across the radio, transport and core networks. By reserving resources, a high availability of time-critical services can be ensured and latencies for queuing can be avoided.

Network orchestration automates the creation, modification and deletion of slices according to a slice service requirement [2]. This can imply that compute locations are selected according to guaranteed resource availability and transport latency rather than the lowest compute costs, for example. 5G New Radio (NR) provides several capabilities for ultra-reliable low-latency communication (URLLC) [7, 9]. From the first

NR standard release, the target has been to enable one-way latencies through the RAN of down to 1ms, where a timely data delivery can be ensured with 99.999 percent probability.

Features addressing low latency include ultra-short transmissions, instant transmission mechanisms to minimize the waiting time for uplink (UL) data, rapid retransmission protocols that minimize feedback delays from a receiver to the transmitter, instant preemption and prioritization mechanisms, interruption-free mobility and fast processing capabilities of devices and base stations.

Features addressing high reliability include a range of robust signal transmission formats. There are methods for duplicate transmissions to improve reliability through diversity, both within a carrier using transmissions through multiple antenna points, as well as between carriers through either carrier aggregation (CA) or multi-connectivity.

Advanced antenna systems (AAS) have tremendous potential to improve the link budget and reduce interference. The vendor-specific radio network configuration, algorithms for scheduling, link adaptation, admission and load control that are at the heart of NR make it possible to fulfill service requirements while ensuring an optimized utilization of available resources.

Support for highly reliable communication has also been addressed for the 5GC, by introducing options for redundant data transmission. Multiple redundant user-plane connections with disjoint routes and nodes can be established simultaneously. This may include the usage of separate user equipment (UE) on the different routes. 5G provides QoS, and by configuring a suitable QoS flow through the 5G system for transporting time-critical communication, queuing latencies due to conflicting traffic can be avoided by traffic separation with resource reservations and/or traffic prioritization.

For a time-critical communication service that is requested by a consumer, a data session

with a suitable QoS flow profile is established, according to a corresponding service subscription. Larger customers, like an enterprise, are typically interested in connectivity for an entire device group. For this purpose, 5G has defined non-public networks (NPNs), which are real or virtual networks that are restricted for usage by an authorized group of devices for their private communication [10]. An NPN can be realized as a standalone network not coupled to a public network that is purpose-built to provide customer services at the customer premises.

Alternatively, an NPN may share parts of the network infrastructure with a public network, like a common RAN that is shared for private and public users. Beyond the shared RAN, the NPN may have a separate dedicated core network and local breakout – that is, it may be located on the customer's premises with its own device authentication, service handling and traffic management. Finally, an NPN can be a network service that is provided by a mobile network operator (MNO) as a customer-specific network slice.

Some NPNs may be customized to provide dedicated functionality for industrial automation, including 5G-LAN services and Ethernet support, providing ultra-low deterministic latency, interworking with IEEE (the Institute of Electrical and Electronics Engineers) TSN, and time-synchronization to synchronize devices over 5G to a reference time [11, 12]. Enhanced service exposure of the 5G system makes it possible to better integrate 5G into an industrial system [13] by means of service interfaces for device management (device onboarding, connectivity management and monitoring, for example) and network management.

... [AAS] HAVE TREMENDOUS POTENTIAL TO IMPROVE THE LINK BUDGET AND REDUCE INTERFERENCE

Spectrum option	Frequency allocation	Deployment scenario	Subcarrier spacing
Low-band FDD	2x10MHz @ 800MHz	Wide area	15kHz
Mid-band FDD	2x20MHz @ 2GHz	Wide area	15kHz
Mid-band TDD	50MHz @ 3.5GHz	Wide area	30kHz
Mid-band TDD	100MHz @ 3.5GHz	Local factory	30kHz
High-band TDD	400MHz @ 30GHz	Local factory	120kHz

Table 1 Spectrum assets considered in the case studies

5G spectrum flexibility

5G NR allows MNOs to take full advantage of all available spectrum assets. NR can be deployed using the spectrum assets used for the LTE networks, either through refarming or spectrum sharing [14]. Most of the LTE spectrum assets are in the low and mid bands, which in the 5G era will continue to be used for wide-area coverage. Traffic growth will drive the need for increased network capacity throughout the 5G era.

Increased capacity can be achieved by adding more spectrum assets, densifying the network and/or upgrading capabilities at existing sites. New 5G spectrum options in the mid bands (around 3.5GHz) and in the high bands (such as the millimeter wave frequencies) present great opportunities with large bandwidths.

Operating with these new spectrum assets, the added RAN nodes can also use advanced hardware features such as an AAS to fully capitalize on the benefits of NR. The coverage provided by the low-band and mid-band spectrum assets is key to enable Critical IoT services in wide-area deployments. Adding network capacity over time will not only increase the capacity for eMBB, but also boost the capacity for Critical IoT.

Case studies

To illustrate how 5G spectrum assets can be utilized for Critical IoT, we have put together case studies for two deployment scenarios: wide-area deployment and local-area deployment inside a factory.

The wide-area scenario is based on a macro-deployment in central London with an inter-site distance of approximately 450m, assuming low-band FDD, mid-band FDD and mid-band TDD spectrum options. For the mid-band deployments, we include an AAS, with eight antenna columns for 3.5GHz and four for 2GHz. Devices with four receiver branches are used in the evaluation.

The local factory setup is based on a factory automation scenario [15] and assumes mid-band and high-band TDD options. Table 1 lists the spectrum options chosen in the case studies.

The top half of Figure 3 presents the served capacity per cell versus various reliability and round-trip RAN latency requirements for outdoor UEs in the central London wide-area deployment scenario. All the TDD cases assume a TDD pattern with 3:1 downlink (DL) and UL split. Observe the cost in terms of capacity when pushing for tighter reliability and latency requirements. Generally, a tighter reliability or latency requirement leads

to a higher consumption of radio resources, as the scheduler needs to provision a larger link adaptation margin to reduce the likelihood of failures in the initial transmissions. Furthermore, we observe that the mid-band options can offer a significant capacity boost for the wide-area scenario, thanks to large available bandwidths and use of AAS.

Among the two mid-band options studied, FDD at 2GHz is attractive when greater UL coverage (99 percent) is desired. Our case studies also show that it is challenging for the wide-area deployment to provide full indoor UL Critical IoT coverage using mid-band spectrum options, due to building-penetration loss. In general, indoor coverage depends on building materials and building sizes.

Under favorable conditions, such as low-loss facades and limited building sizes (that is, less than 3,600sq m in footprint), it is feasible

to have 95 percent indoor UL coverage even using the mid-band carriers, although the achievable capacity is limited. Local indoor deployments are a prerequisite in high-loss or very large buildings, and are also necessary in other buildings if high indoor coverage and capacity is desired.

Although suburban and rural scenarios typically have larger cells, it is nonetheless possible to achieve similar results there. This is because antennas in suburban and rural environments tend to be installed at a greater height, there are fewer obstacles and the smaller buildings result in less wall-penetration loss. These factors compensate for the differences in cell range, making it feasible to achieve very good Critical IoT performance in suburban and rural scenarios as well.

For local-area studies (scenario #2), the deployment using 3.5GHz spectrum is based

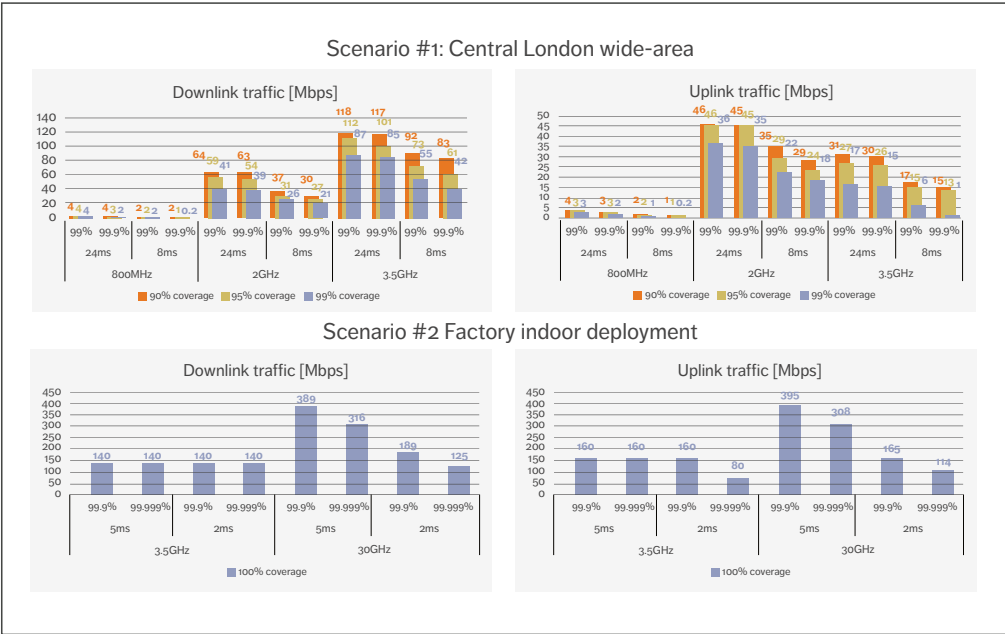


Figure 3 Served capacity per cell versus various reliability and round-trip RAN latency requirements for the two scenarios

●● MNOS CAN START TO ADDRESS TIME-CRITICAL USE CASES ... THROUGH SOFTWARE UPGRADES ●●

on a single-cell deployment with eight antennas installed in the ceiling, uniformly distributed across the entire factory, and a DL and UL symmetric TDD pattern. For the high-band deployment, eight transmission points with full frequency reuse are considered. The 3GPP indoor factory channel model with dense clusters, including machinery, assembly lines, storage shelves and so on [16], is used. To achieve 2ms round-trip RAN latency, NR mini-slot and configured grant features are used. (Using the same features, an FDD carrier with 15kHz subcarrier spacing can also achieve similar latency.)

The bottom half of Figure 3 shows that both DL and UL traffic achieve 100 percent coverage. Tightening Critical IoT requirements reduces capacity, however, and this is more evident in the high-band case. For the mid-band case, all users consistently reach the highest spectral efficiency except for UL traffic with the most stringent requirements, due to good coverage and the absence of interference achieved by the single-cell distributed antenna deployment.

5G NR CA allows radio resources from multiple carriers in multiple bands to be pooled to serve a user. For example, DL traffic can be delivered using a mid-band carrier even when the UL service requirements are not attainable on that mid-band carrier, by using a low-band carrier for the UL control and data traffic. This allows the DL capacity of the mid-band carrier to be utilized to a greater extent.

In essence, inter-band CA allows an MNO to improve coverage, spectral efficiency and capacity by dynamically directing the traffic through the

better carrier, depending on the operating condition, user location and use case requirements.

With a low-band carrier, there are also benefits of pooling an FDD carrier and a TDD carrier from a latency point of view, using the FDD carrier to mitigate the extra alignment delay introduced on a TDD carrier due to the DL-UL pattern.

Deployment strategy

MNOs have started to upgrade some 4G LTE radio base station equipment to 5G NR through software upgrades. The dynamic spectrum sharing solution allows efficient coexistence of LTE and NR in the same spectrum band down to millisecond level [14].

MNOs can start to address time-critical use cases in the wide area (the entertainment, health care and education sectors, for example) by adding support for Critical IoT connectivity to the NR carriers through software upgrades. More stringent, time-critical requirements call for radio network densification, edge computing, and further distribution and duplication of core network functions, which can be done gradually over time, while maximizing returns on investment.

In the confined wide-area scenarios (railways, utilities, public transport and the like), relatively stringent requirements can be addressed with reasonable investments in existing and new infrastructure. In local-area scenarios such as factories, ports and mines, even extreme time-critical requirements can be supported once the E2E ecosystem is established.

Dedicated spectrum has been allocated to some industry sectors in certain regions. In the wide-area scenarios such as public safety and railways, the allocated bandwidths are typically small (10MHz or below) and unable to meet the capacity demands of emerging use cases, especially those with time-critical requirements.

In some regions, significant TDD spectrum has been allocated to enterprises for local use (in the order of 100MHz) in mid-band and millimeter-wave frequency ranges. For both confined wide-area and local-area scenarios, the reuse of MNOs' existing infrastructure and their flexible spectrum assets (in combination with dedicated spectrum, if available) brings major value and opportunities. This approach makes it possible to exploit the full potential of various band combinations and support seamless mobility and interaction between public and dedicated communication infrastructure.

Conclusion

Critical Internet of Things connectivity addresses time-critical communication needs across various industries, enabling innovative services for consumers and enterprises. Mobile network operators are uniquely positioned to enable time-critical services with advanced 5G networks in a systematic and cost-effective way, taking full advantage of flexible spectrum assets, efficient reuse of existing footprint and flexible software-based network design.

Further reading

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Integrated access and backhaul

A NEW TYPE OF WIRELESS BACKHAUL IN 5G

5G New Radio introduces a new type of wireless backhaul known as integrated access and backhaul that is of particular interest for dense deployment of street-level radio nodes.

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The combination of millimeter wave (mmWave) spectrum – which is becoming available globally for 5G – with other spectrum assets below 6GHz results in high speeds and capacities. The mmWave radio resources can only provide limited coverage, though, which makes it reasonable to expect a fairly low level of utilization. As a result, there is an opportunity to use an innovative type of wireless backhaul in 5G – integrated access and backhaul (IAB) – to densify networks with multi-band radio sites at street level.

■ Transport networks play a vital role in RANs by connecting all the pieces. The use of dark fiber for 5G transport is of growing importance [1], and wireless backhaul is an essential complement for sites where fiber is either not available or too costly. In fact, microwave backhaul has been the dominant global backhaul media for over

two decades and will remain a highly attractive complement to fiber for 5G transport [2]. Network densification using street-site deployments comes with new challenges, however. The allowed space and weight for equipment is limited. The installation, integration and operation must be simplified with a high degree of automation to achieve cost-efficient deployment of RAN and transport. This calls for a new type of wireless backhaul that is fully integrated with 5G New Radio (NR) access. This is where IAB enters the frame. More than 10GHz of total bandwidth in the mmWave frequency range of 24.25GHz to 71GHz was globally identified for 5G at the ITU World Radio Conference 2019. Already today, 5GHz of mmWave bandwidth is available in the US. The best overall performance at the lowest total cost of ownership is achieved by using mmWave in combination with spectrum assets below 6GHz [1].

These assets will be deployed on macro sites (rooftops, towers) and street sites (poles, walls, strands) in urban areas with high demands on capacity and speed, as well as in suburban areas with fiber-like fixed wireless access (FWA) services [3]. IAB could provide fast deployment of mmWave backhaul for new multiband street sites, with an easy migration to fiber-based backhaul if, and when, needed.

Using radio-access technology to provide backhaul

Access spectrum has historically been too valuable and limited to use for backhauling. Its rare use today is for LTE solutions that provide a single backhaul hop using a separate frequency band from access, as shown in section A of *Figure 1*. This approach uses a fixed wireless terminal (FWT) to provide connectivity to a separate backhaul core instance. The instance could either be in the core for radio access or distributed closer to the radio nodes to support lower latency inter-site connectivity. It is also possible to use 5G NR to provide such separate access and backhaul solutions.

A solution more like IAB was studied for LTE in 3GPP release 10 in 2011, also known as LTE relaying [4], but it never gained any commercial interest. However, with the wide mmWave bandwidths now becoming available, there is considerable interest in an IAB solution for 5G NR. The work on IAB has been going on in the 3GPP since 2017, and it is currently being standardized for release 16, targeting completion during 2020 [5, 6]. IAB can provide flexible and scalable multi-hop backhauling, using the same or different frequency bands for access and backhaul, as shown in section B of *Figure 1*. The backhaul is efficiently forwarded across the wirelessly interconnected radio nodes, with the backhaul links terminated by an IAB mobile termination (IAB-MT) function. The IAB-MT could either use a separate antenna or share the access antenna of the base station (virtual IAB-MT). The latter provides the ultimate level of integration, as well as utilizing the high-performance base station antennas for backhaul over longer distances.

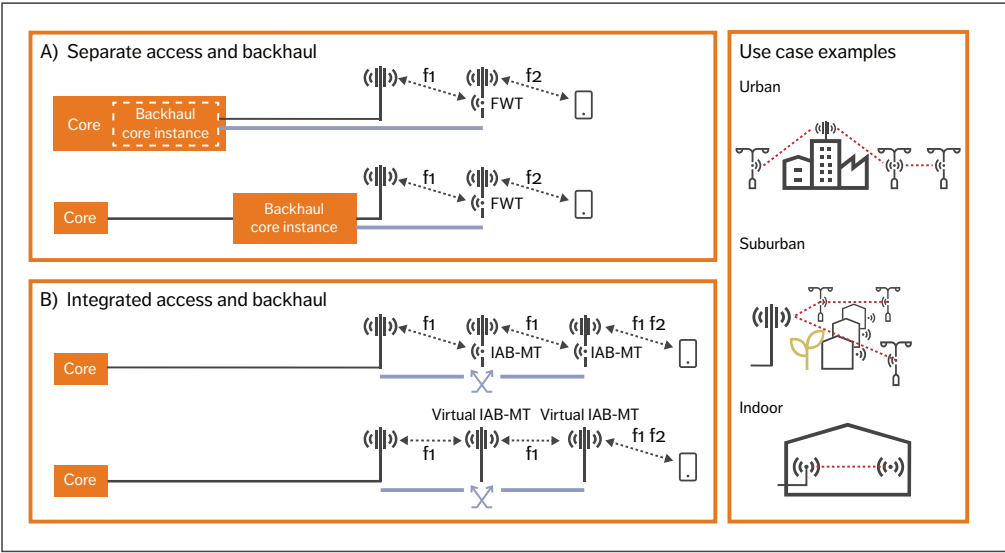


Figure 1 Solutions using radio-access technology to provide backhaul

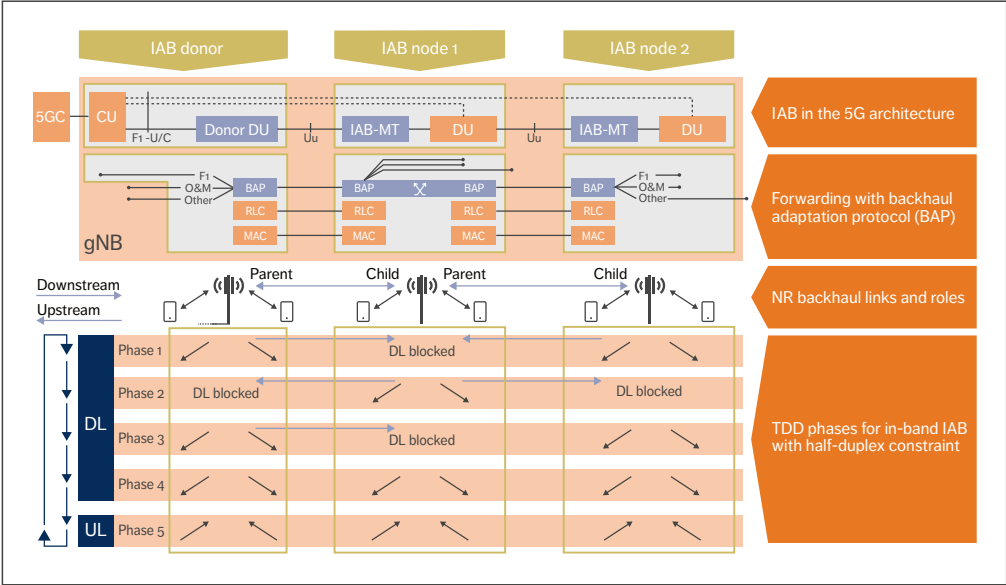


Figure 2 The 3GPP IAB concept

The IAB concept is defined by the 3GPP to be flexible and scalable to support other use cases beyond the initial market interest, such as low-power indoor radio nodes. There is also research on future advanced enhancement and optimizations for more visionary IAB use.

The 3GPP concept of integrated access and backhaul

IAB is defined to reuse existing 5G NR functions and interfaces, as well as to minimize impact on the core network. The architecture is scalable, so that the number of backhaul hops is only limited by network performance. From a transport perspective, IAB provides generic IP connectivity to enable an easy upgrade to fiber transport when needed.

In the 5G network, the gNB base station provides NR protocol terminations to the user equipment (UE) and is connected to the 5G Core (5GC) network. As defined in 3GPP TS 38.401 [7], the gNB is a logical node, which may be split into one central

unit (CU) and one or more distributed units (DU). The CU hosts the higher layer protocols to the UE and terminates the control plane and user plane interfaces to the 5GC. The CU controls the DU nodes over the F1 interface(s), where the DU node hosts the lower layers for the NR Uu interface to the UE.

As illustrated in Figure 2, the CU/DU split architecture is used for IAB and enables efficient multi-hop support. The architecture eliminates the backhaul core instance at every IAB node shown in Figure 1A and related overhead due to tunnels inside tunnels, which would become severe for large multi-hop chains.

As the time-critical functionality is located in each DU, the F1 interface is well suited for a non-ideal backhaul such as IAB. The IAB donor is a logical node that provides the NR-based wireless backhaul and consists of a CU and wire-connected donor DU(s). The IAB nodes, which may serve multiple radio sectors, are wireless backhauled to the IAB donor and consist of a DU and an IAB-MT.

All IAB-nodes and donor DU(s) that use the same CU are part of one gNB, in accordance with the CU/DU split architecture. Hence, the wireless backhaul is isolated inside the gNB, and any internal topology, routing or backhaul changes can be made without impacting the 5GC or neighboring gNBs. A similar situation is valid for the UEs, for which the IAB node appears as a normal base station, supporting both NR standalone and non-standalone mode.

As shown in Figure 2, the NR backhaul link is between a “parent” on the network side and a “child” at the other end. The DU at the parent schedules the backhaul downstream and upstream traffic to/from the IAB-MT at the child, supporting a limited subset of the NR UE functionality. This includes lower protocol layer functionality to the parent as well as Radio Resource Control and non-access stratum functionality to the IAB donor CU and 5GC.

The backhaul adaptation protocol (BAP) [11] enables efficient IP data forwarding across the IAB interconnected radio nodes, where the BAP data is carried by backhaul Radio Link Control (RLC) channels on each NR backhaul link. Multiple channels can be configured to enable traffic prioritization and QoS enforcement and, based on these properties, the BAP entity in each node maps protocol data units to the appropriate backhaul RLC channel.

Hop-by-hop forwarding, from the IAB donor to the destination IAB node, is based on the BAP routing identity set by the IAB donor. Any IP traffic can be forwarded over the BAP, such as F1 and operation and maintenance (O&M) of the IAB nodes, as well as connectivity of any other equipment at the IAB-node site, as shown in Figure 2.

Physical layer aspects

The IAB feature is intended to support out-of-band and in-band backhauling, where the latter means usage of the same carrier frequencies for both the NR backhaul links and the access links. In-band operation comes with a half-duplex constraint, implying that the IAB-MT part of an IAB node cannot receive while its collocated DU is transmitting and vice versa to avoid intra-site interference. A strict time-domain separation

is therefore required between transmission and reception phases within each IAB node.

IAB is expected to be of most benefit in mmWave spectrum, where TDD [8] is used and operators typically have large bandwidth. A TDD network is typically configured with a (often regulated) pattern for the time domain allocation of downlink (DL) and uplink (UL) resources, and an additional level of pattern must be used to support combined access and backhaul traffic. This is illustrated in the example in Figure 2, with five different repeated IAB time phases for node-local TDD states, where phases 1-4 are mapped to the DL and phase 5 to the UL.

IAB IS EXPECTED TO BE OF MOST BENEFIT IN MMWAVE SPECTRUM

The mix and duration of different phases can be flexible depending on the scenario, access/backhaul link performance, load and so on. Due to the half-duplex constraint, there will be time periods in which the nodes are blocked from transmission in a normal DL slot, effectively reducing the peak rate for an IAB node compared with a similar node with wired (non-limited) backhaul. This occurs whenever there is a transmission over the NR backhaul link, as the receiving end of the link will not operate according to the overall TDD pattern. In the example in Figure 2, the backhaul transmission occurs in phases 1-3, and the normal DL operation is blocked for the receiving nodes (in all sectors) during these phases.

The parent node schedules all traffic over the backhaul link (phases 1-3) in the same way as for UE scheduling, where frequency division multiplexing or space division multiplexing can be used to separate simultaneous transmissions.

Deployment constraints for integrated access and backhaul

From a 3GPP architecture perspective, the IAB feature is flexible, supporting multi-hop and a variety of topologies. However, there are other aspects that

restrict the size of the IAB network topology, where in-band operation (sharing spectrum for both backhaul and access) is an essential reason for these limitations. Larger IAB topologies might also require complex control functions. But since IAB is a complement to fiber, the size of most IAB networks is expected to be small.

In a multi-hop network, the first backhaul hop must carry the backhaul bandwidth not only for the first IAB node, but also for all other IAB nodes further down in the hop chain. Deploying multi-hop networks will therefore eventually lead to backhaul-limited nodes due to congestion in the first hop. Increasing the number of hops will also increase the end-to-end latency and raise the complexity for scheduling and routing to satisfy QoS.

The 3GPP gNB synchronization requirements apply also for IAB nodes that may be fulfilled with a node-local synchronization solution based on the Global Navigation Satellite System. In some situations, this is neither wanted nor feasible. Over-the-air synchronization is therefore an alternative option, using periodic parent-transmitted reference symbols as the synchronization source for the receiving child node. This scheme implies that the clock accuracy at the donor DU must be better than the 3GPP requirement, as the synchronization budget is shared/ aggregated for all nodes using this donor DU. There are therefore several practical reasons to limit the number of hops and not deploy oversized IAB topologies.

Regardless of topology, there are also general radio aspects to consider. The 3GPP specifies radio interface requirements for the IAB-MT [8], with two categories to distinguish different use cases and characteristics. One category is for wide-area usage with planned site deployment, such as backhaul of street sites; the other is for local-area usage with site deployments that may not be preplanned.

The wide-area category enables an integrated solution for the access and backhaul links, where the IAB node can benefit from using the full base station capabilities – such as advanced antenna systems (AAS) [9] and high output power – to provide good backhaul link performance and relatively large distance between parent and child nodes.

Wide-area and local-area IAB-MT are intended for different deployment scenarios and use differing TDD patterns. In Figure 2, all backhaul link traffic is scheduled during DL timeslots but an alternative TDD scheme may be applied where the UL timeslots are used for upstream backhaul. The latter scheme is restricted in terms of output power, making it more suitable for local-area deployments.

The IAB backhaul links give rise to a semi-synchronous TDD operation, for which the regulatory framework for local coordination between operators is not yet in place in all countries [10]. As illustrated by the TDD phases in Figure 2, during certain time slots the IAB node will operate in an inverted mode with respect to the general TDD pattern. This means that a node may be in receiving mode during a DL slot for backhaul link reception and thus suffer from neighbor node interference, both within the same channel as well as between channels in the same frequency band.

Even though the backhaul link is more robust against interference due to good link budget, measures such as isolation between nodes (separation distance, for example) or coordinated TDD patterns may still be required to avoid excessive interference.

THE SIZE OF MOST IAB NETWORKS IS EXPECTED TO BE SMALL

Integrated access and backhaul from a backhaul perspective
Traditional backhaul is a service provided by the transport network domain to the radio-access nodes. For IAB, a segment of the backhaul is embedded in the RAN domain, sharing common radio resources. The backhaul transport cannot be dimensioned on an individual node basis, as the IAB donor terminates the “common backhaul” for all underlying IAB nodes extending the radio access to UEs through a network of backhaul and access links.

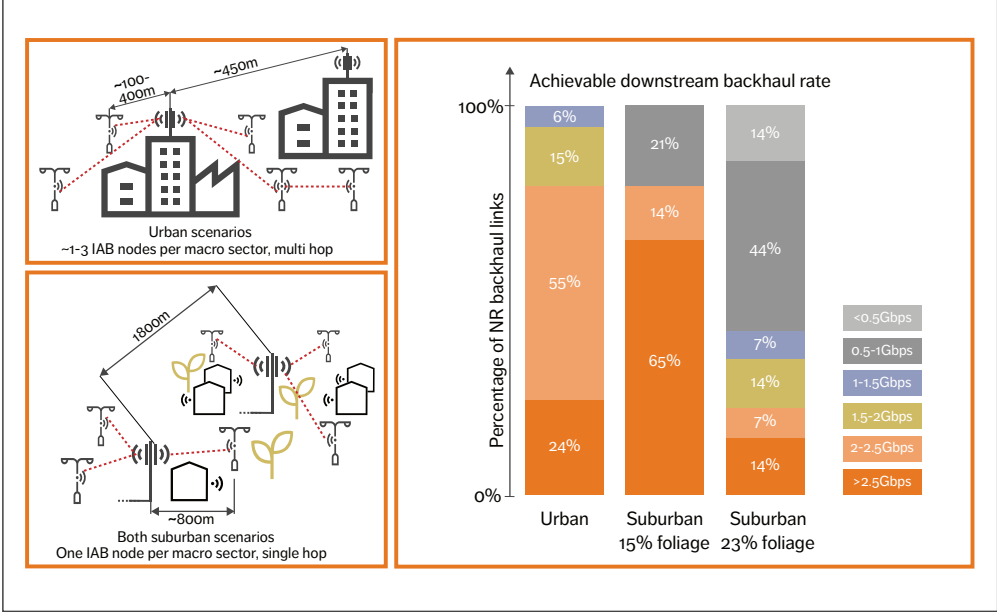


Figure 3 Simulated IAB performance for three scenarios

Instead, the backhaul dimensioning for IAB systems needs to be an integrated part of RAN dimensioning, considering the shared radio resources for backhaul and access. From a transport network perspective, the IAB nodes appear as extensions of the IAB donor. The same IP assignment methods can be used for IAB nodes as for fiber-connected radio nodes, which facilitate an easy upgrade to fiber transport when needed.

IAB can also provide IP connectivity for other equipment at the IAB node site, as shown in Figure 2. The transport performance requirements to the IAB donor are affected by the connected IAB nodes. The busy hour data traffic is funneled through the IAB donor and increases with each connected IAB node. The latency and synchronization requirements for the transport are also affected, as each IAB backhaul hop adds latency and timing error. These aspects will also limit the size of the IAB topology.

The role of integrated access and backhaul in network evolution

Densification of current networks will mainly take place in urban and dense suburban environments. As one part of assessing the role of IAB, we have performed radio network simulations of such scenarios, as illustrated in Figure 3.

When providing broadband to a home, FWA is a good alternative to fiber in many cases, as it lowers the barrier to entry and supports faster deployment [3]. In cases where the traffic demands require densification – in dense suburban areas in the US, for example – the use of wireless backhaul can further add to these advantages.

We studied FWA using IAB in simulations of two US suburban neighborhoods in the San Francisco Bay Area, with the level of foliage as the main difference: 15 and 23 percent respectively. As a reference, our estimates indicate that about half of the dense suburban areas in the US have a foliage level lower than 15 percent.

Both areas had a macro grid, with an inter-site distance of 1,800m and three sectors per site. In order to serve households with an average data consumption of 1,000GB per month, a densification was made where each macro sector provided backhaul to a street site with a single hop of about 800m, as shown in Figure 3.

All sites were deployed with 40MHz on mid band for access and 800MHz on mmWave for access and backhaul. The locations of the new street sites were chosen to secure good backhaul links from the macro sites as well as good access coverage to the homes. An ideal position can be a utility pole in line-of-sight (LOS) of the macro site with a surrounding area with few or no obstacles. Moreover, the risk of future infrastructure changes blocking the backhaul link should be considered.

In the area with less foliage, the street sites off-load the macro sites by serving around 40 percent of the households. About 80 percent of the backhaul links have a downstream rate above 2Gbps, as shown in Figure 3. Over 200 households per square kilometer could be served without IAB causing any traffic limitation, even during peak hours.

In the area with more foliage, the propagation conditions are worse both for access and backhaul. Therefore, additional street sites may be needed to meet the required capacity, which will affect the business case. It was more challenging to find street site locations with good access as well as backhaul. Around 60 percent of the backhaul links have a downstream rate below 1Gbps, which means the backhaul will consume a large part of the common

access and backhaul radio resources. Fiber backhaul is therefore recommended for such sites. For the remaining 40 percent of the sites, IAB could be a viable option, despite the amount of foliage.

In a simulation of urban London, a densification with street sites is required to extend coverage and improve mobile broadband capacity both indoors and outdoors. All sites use mid band and mmWave for access, and the mmWave is also used for backhauling between street sites and macro sites. The backhaul topology was a tree structure with one to four hops, where most sites only had a single hop.

The simulations show that the need for an LOS backhaul link is less critical in urban environments than in suburban ones, thanks to strong reflections in the city environment, making it relatively easy to find locations with good signal strength.

Furthermore, the backhaul links are shorter in an urban environment, and the impact of foliage is typically less significant due to fewer trees. Figure 3 shows the achievable downstream backhaul link rates for the urban case, which are all above 1Gbps. Eighty percent are above 2Gbps. The densified network provides excellent coverage and capacity for both outdoor and indoor users, even though IAB consumes part of the spectrum.

For both suburban and urban scenarios, these simulations show that IAB is an attractive complement to fiber, with the ability to provide backhaul in the early years until traffic growth requires all radio resources to be used for access. Depending on the subscriber distribution, IAB may not even need to be replaced by fiber at some sites.

Terms and abbreviations

5GC – 5G Core | **AAS** – Advanced Antenna System | **BAP** – Backhaul Adaptation Protocol | **CU** – Central Unit | **DL** – Downlink | **DU** – Distributed Unit | **F1** – Interface CU–DU | **FWA** – Fixed Wireless Access | **FWT** – Fixed Wireless Terminal | **gNB** – gNodeB | **IAB** – Integrated Access and Backhaul | **LOS** – Line-of-Sight | **MAC** – Medium Access Control | **mmWave** – Millimeter Wave | **MT** – Mobile Termination | **NR** – New Radio | **O&M** – Operation and Maintenance | **PoC** – Proof of Concept | **RLC** – Radio Link Control | **TDD** – Time Division Duplex | **UE** – User Equipment | **UL** – Uplink | **Uu** – radio interface between RAN and UE

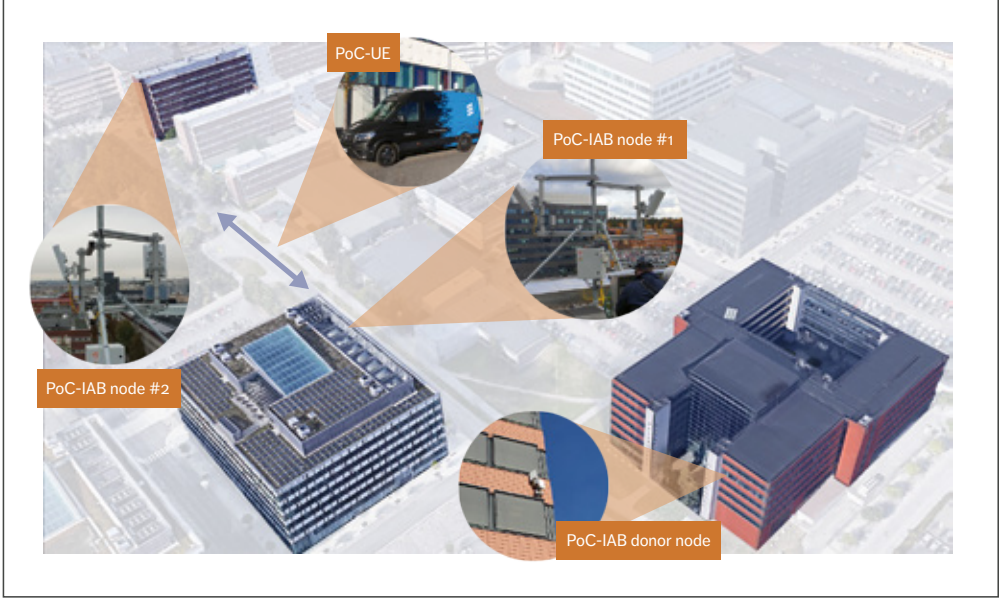


Figure 4 Deployment of the IAB testbed in Stockholm

As the backhaul link uses parts of the available spectrum resources, the typical rates for users in cells served by donors or IAB nodes will be lower than if all nodes are fiber connected. Still, with the small scale network topologies used in the simulations, we achieve peak user throughputs far above 1Gbps.

Proof of concept

In order to properly assess IAB-specific performance aspects, Ericsson has developed an IAB proof of concept (PoC) testbed in an authentic environment. At the Ericsson site in Stockholm we have set up a two-hop IAB deployment using two-sector IAB nodes with either 28GHz or 39GHz AAS radios with 100MHz bandwidth, as shown in Figure 4.

The IAB nodes are in the wide-area category with NR backhaul links using DL slots for all transmitted data. Initial test results are aligned with expectations and show backhaul bitrates near the predicted and end-to-end peak rates independent of hop level.

Conclusion

The massive amount of mmWave spectrum that is becoming available globally will spark a wide variety of innovative 5G use cases. Integrated access and backhaul (IAB) is one such innovation that could enhance 5G New Radio to support not only access but also wireless backhaul.

Our radio network simulations show that IAB could serve as a versatile backhaul option for street sites in urban and suburban areas, using small-scale star and tree backhaul topologies. It could also be useful for temporary deployments for special events or emergency situations.

Point-to-point microwave backhaul will remain an essential complement to fiber for 5G transport for traditional macro sites, while IAB is a promising advanced concept that may become as important for wireless backhaul of street sites.

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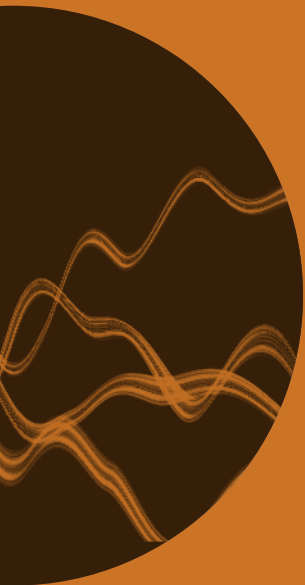
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